

Exact POMDP Solutions: α -vectors

Recap

Recap

- POMDP

Recap

- POMDP

$$(S, A, O, R, T, Z, \gamma)$$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

$$b_t(s) = P(s_t = s \mid h_t)$$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

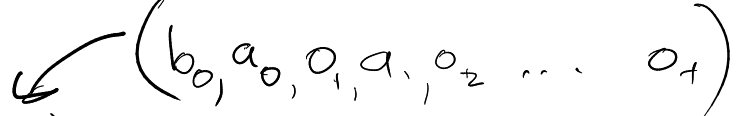
$$b_t(s) = P(s_t = s \mid h_t)$$

$$b' = \tau(b, a, o)$$

Recap

- POMDP $(S, A, O, R, T, Z, \gamma)$
- Belief Updates

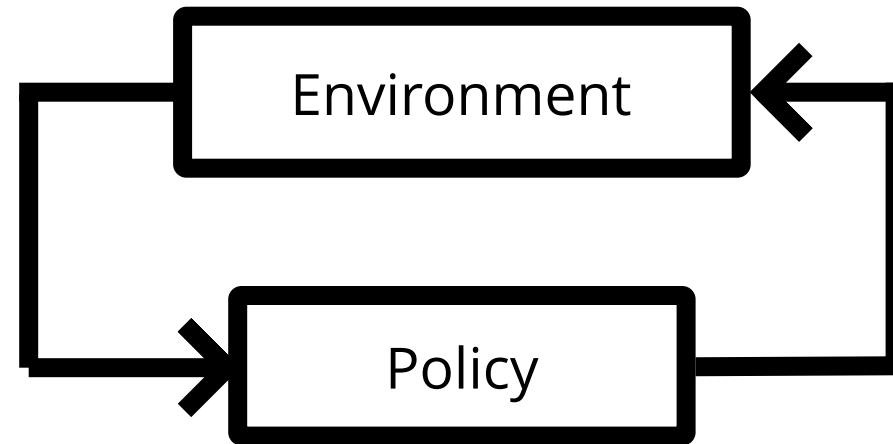
$$b_t(s) = P(s_t = s \mid h_t)$$


A handwritten tuple $(b_0, a_0, o_1, a_1, o_2, \dots, o_t)$ is shown. A curved arrow points from the o_1 term to the h_t term in the equation above, indicating that the history h_t contains the sequence of past actions and observations.

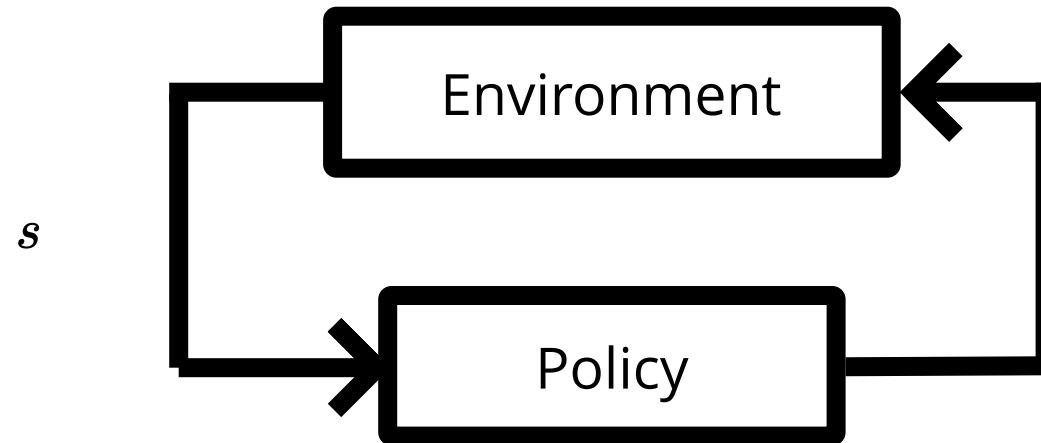
$$b' = \tau(b, a, o)$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

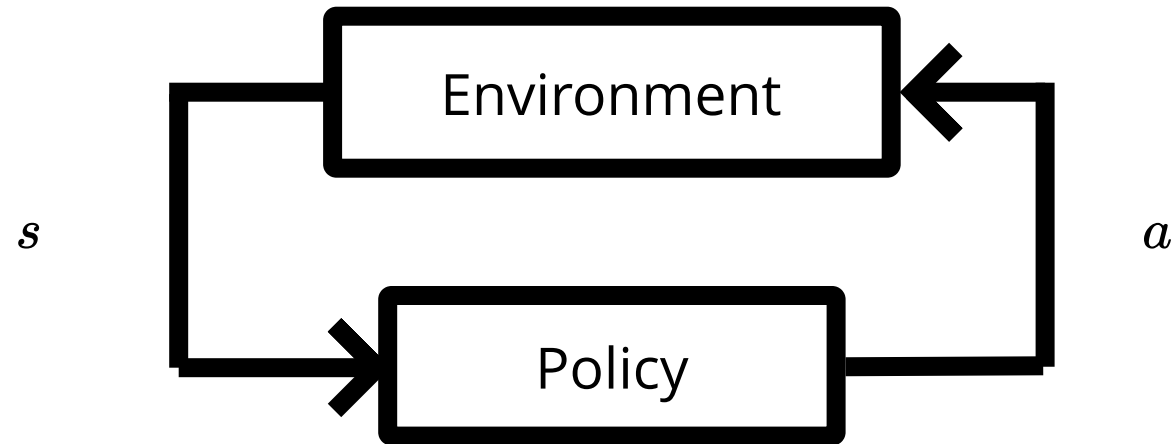
MDP Sense-Plan-Act Loop



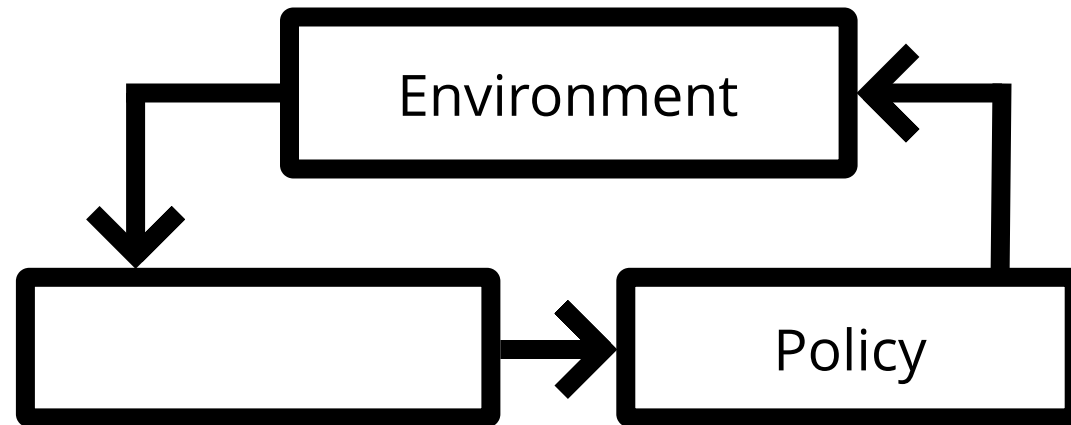
MDP Sense-Plan-Act Loop



MDP Sense-Plan-Act Loop



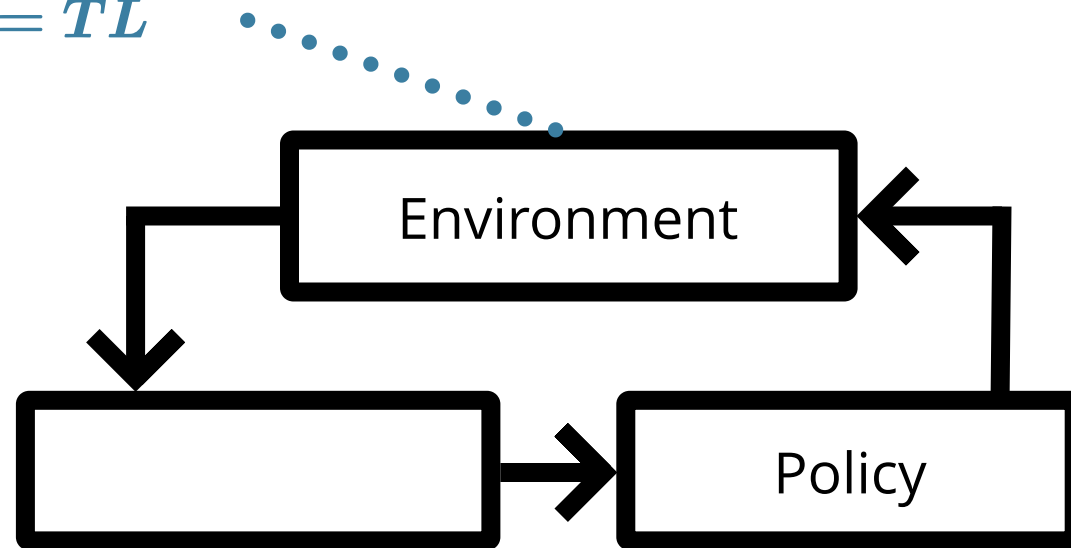
POMDP Sense-Plan-Act Loop



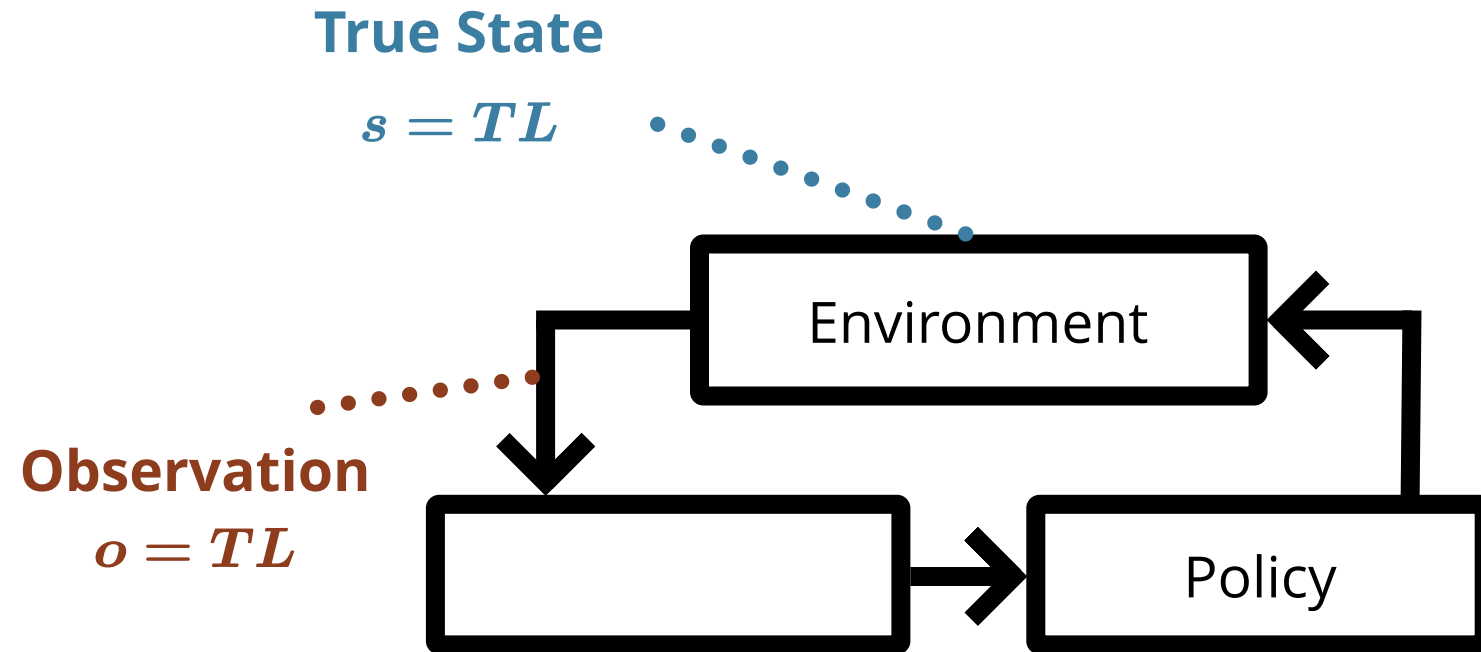
POMDP Sense-Plan-Act Loop

True State

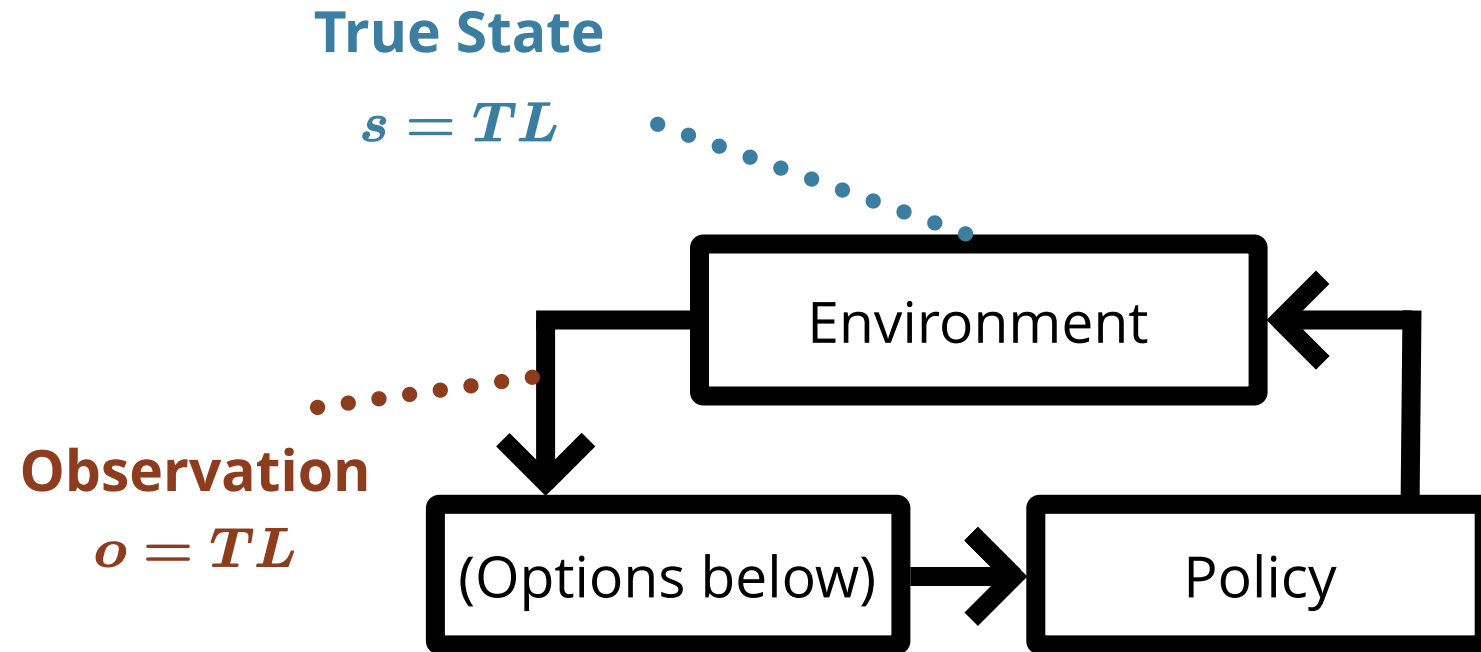
$$s = TL$$



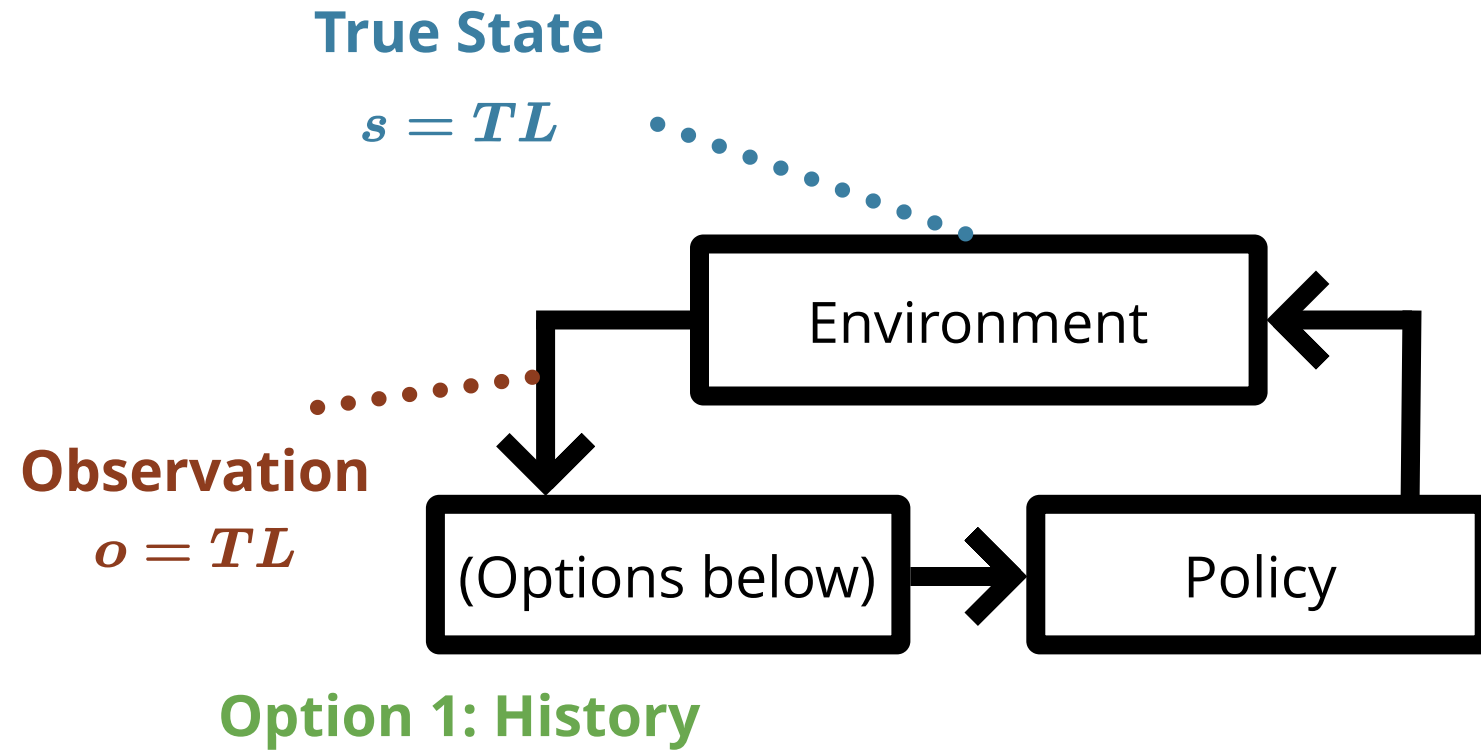
POMDP Sense-Plan-Act Loop



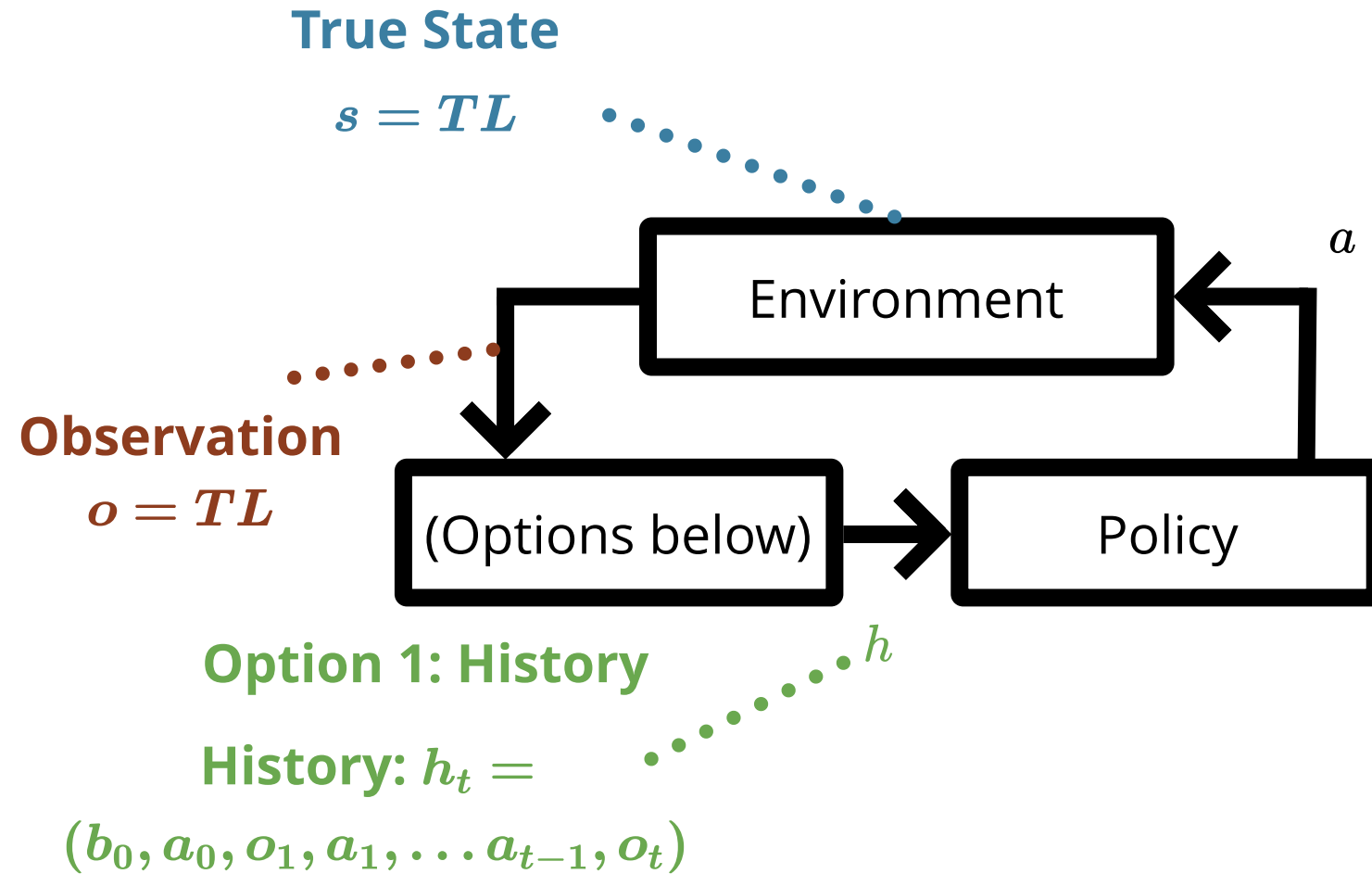
POMDP Sense-Plan-Act Loop



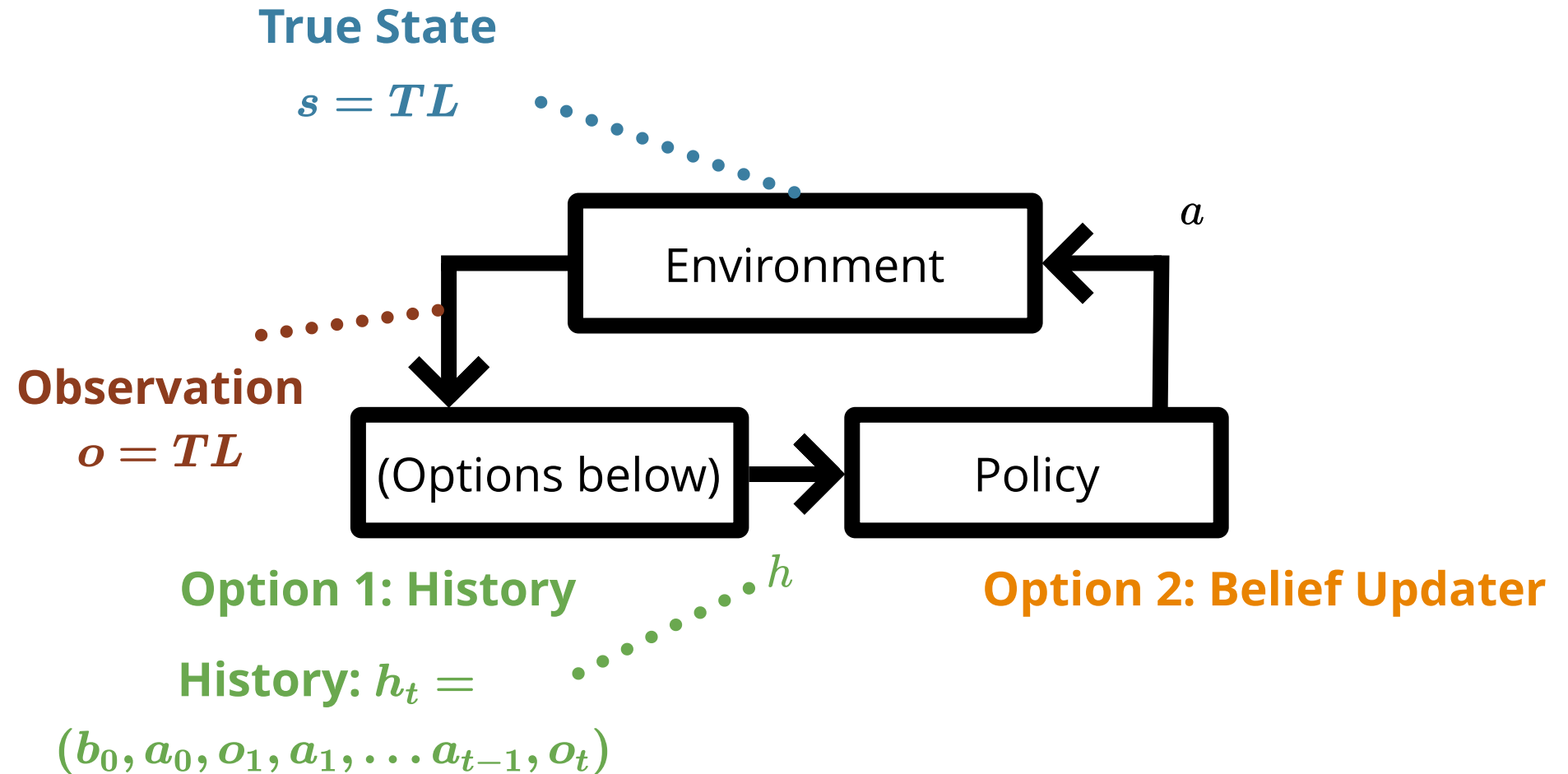
POMDP Sense-Plan-Act Loop



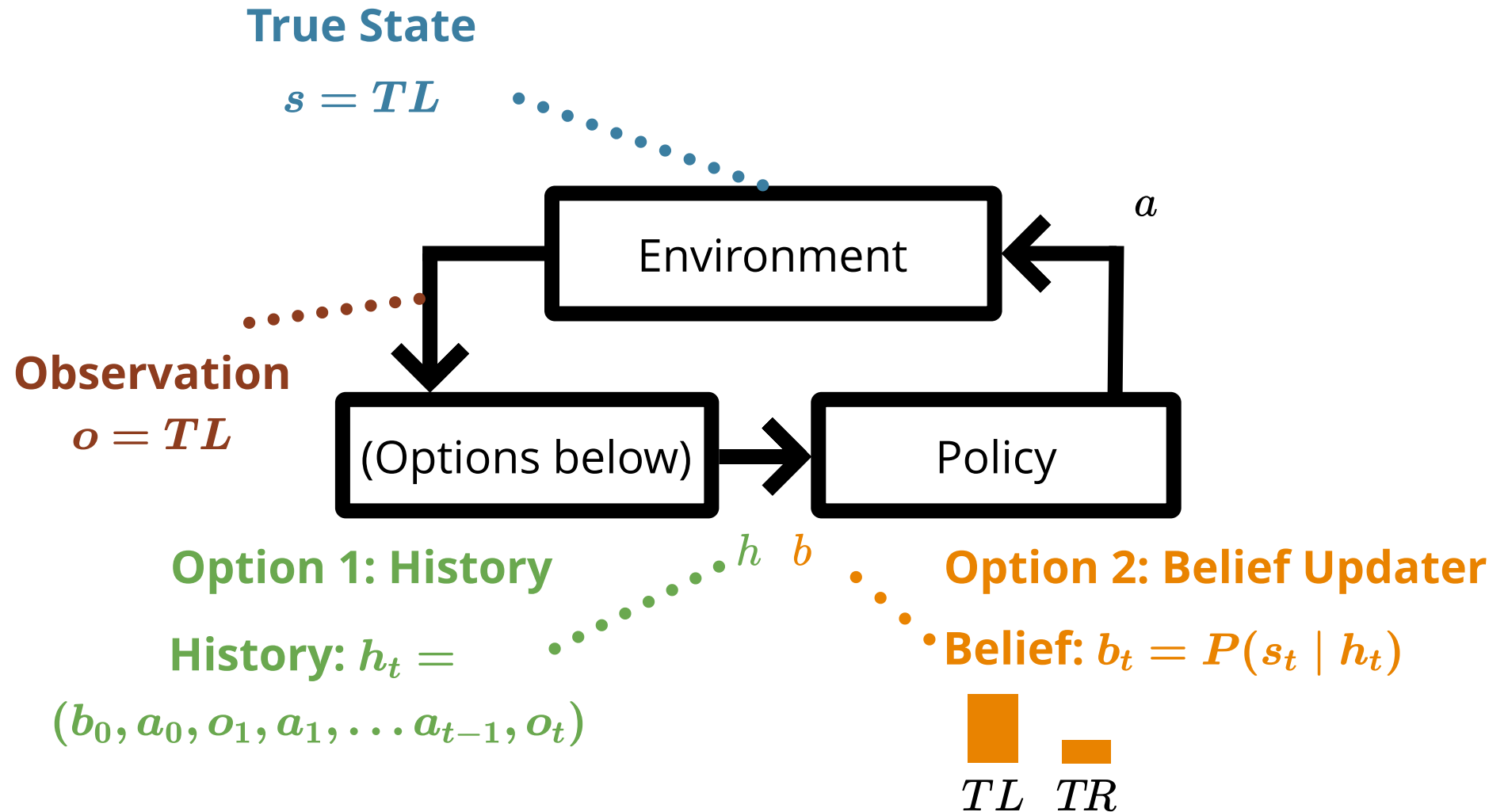
POMDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop



POMDP Sense-Plan-Act Loop



Exercise 1: Crying Baby Belief Update

Exercise 1: Crying Baby Belief Update

$$S = \{h, \overset{\text{not}}{\neg}h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

$$R(s, a) = R(s) + R(a)$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$A = \{f, \neg f\}$$

$$O = \{c, \neg c\}$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

Exercise 1: Crying Baby Belief Update

$$S = \{h, \neg h\}$$

$$T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\}$$

$$T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\}$$

$$T(\neg h \mid \cdot, f) = 1.0$$

$$b'(s') \propto \underbrace{Z(o \mid a, s')} \sum_s T(s' \mid s, a) b(s)$$

Starting at a $b(h) = 0$, calculate

b' with $a = \neg f$ and $o = c$.

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

$$b'(h) \propto Z(c \mid \neg f, h) (T(h \mid h, \neg f) b(h) + T(h \mid \neg h, \neg f) b(\neg h))$$

$$0.8 \quad (1.0 \quad 0.0 + 0.1 \quad 1.0)$$

$$\rightarrow b'(h) \propto 0.08$$

$$b'(\neg h) \propto Z(c \mid \neg f, \neg h) (T(\neg h \mid h, \neg f) b(h) + T(\neg h \mid \neg h, \neg f) b(\neg h))$$

$$0.1 \quad (0 \quad 0.9 \quad 1.0)$$

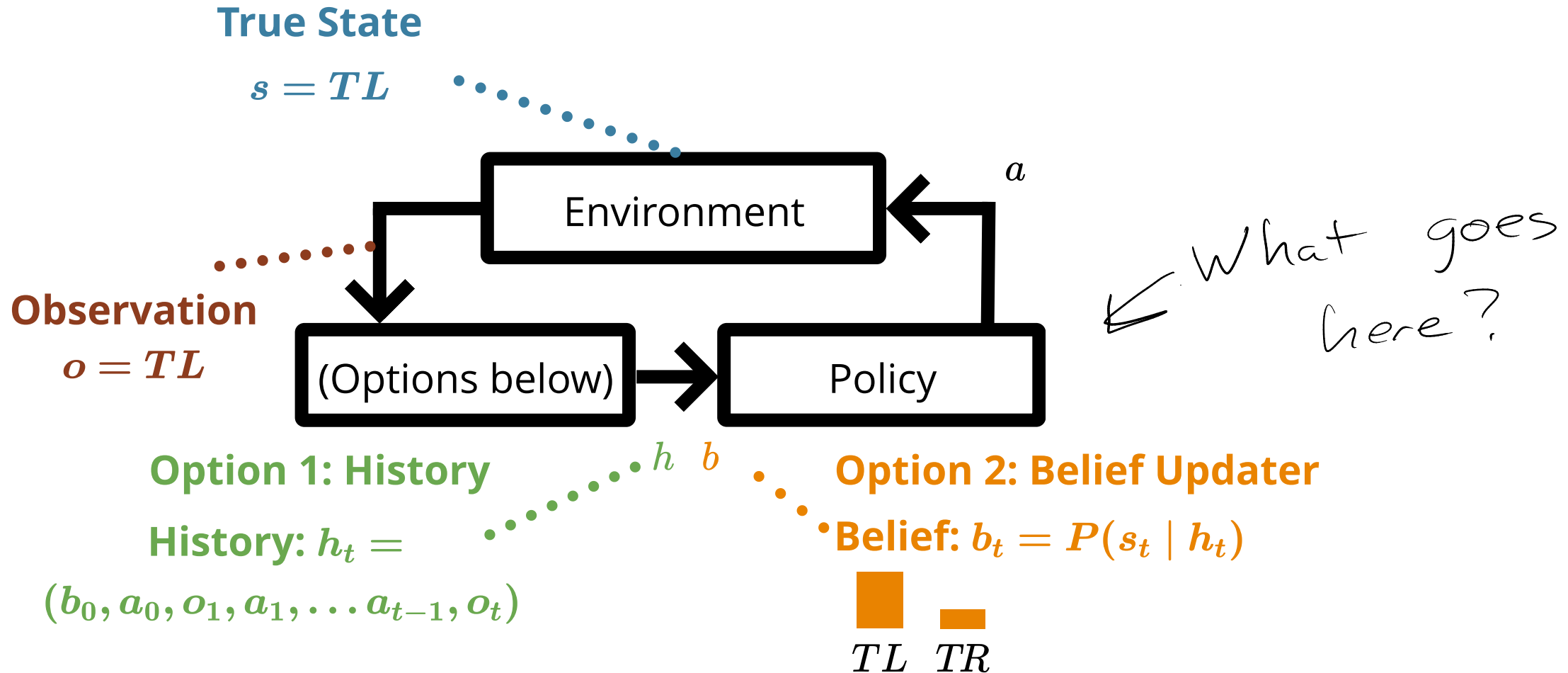
$$\rightarrow b'(\neg h) \propto 0.09$$

$$b'(h) = \frac{0.08}{0.08 + 0.09} = 47\%$$

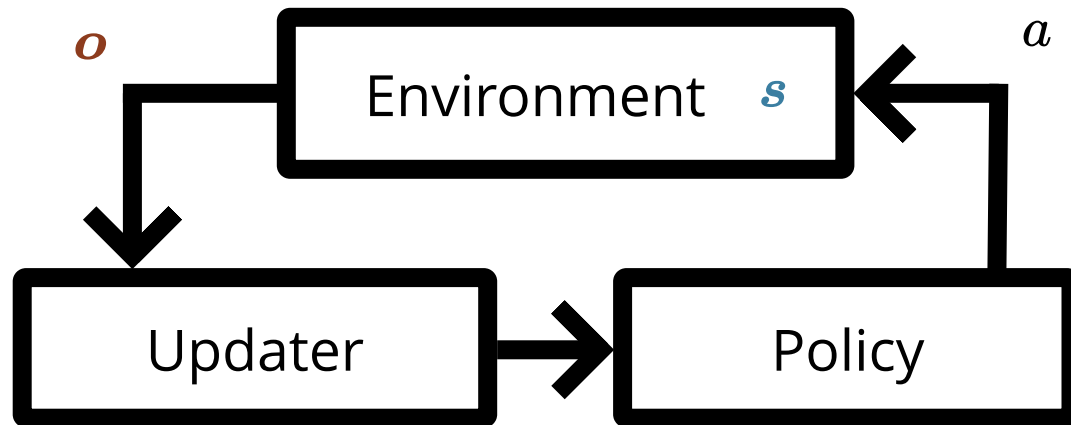
$$b'(\neg h) = 1 - 47\% = 53\% = \frac{0.09}{0.08 + 0.09}$$

Guiding Question

How do we calculate the optimal action in a POMDP?



Belief-Space MDP

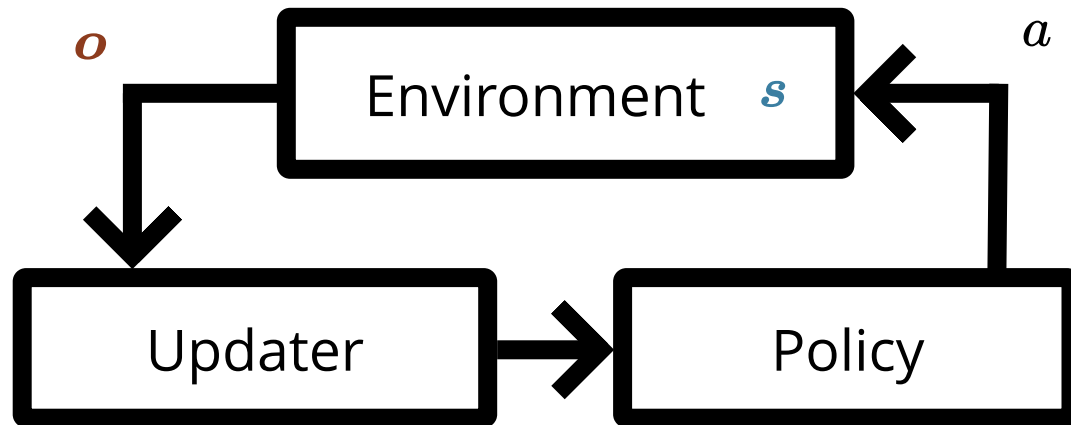


$$b_t = P(s_t | h_t)$$


TL


TR

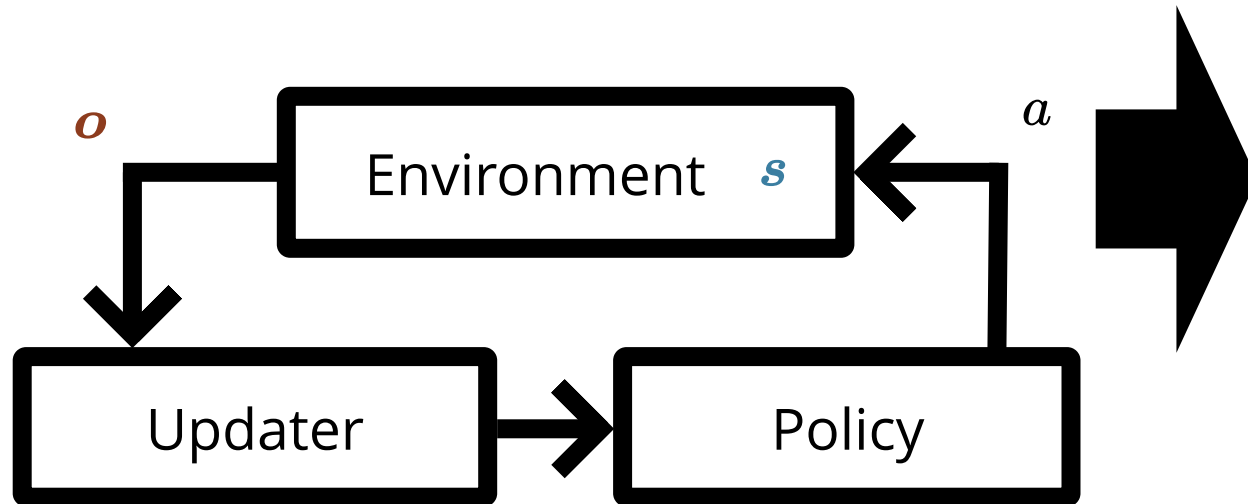
Belief-Space MDP



$$b_t = P(s_t | h_t)$$

 TL  TR

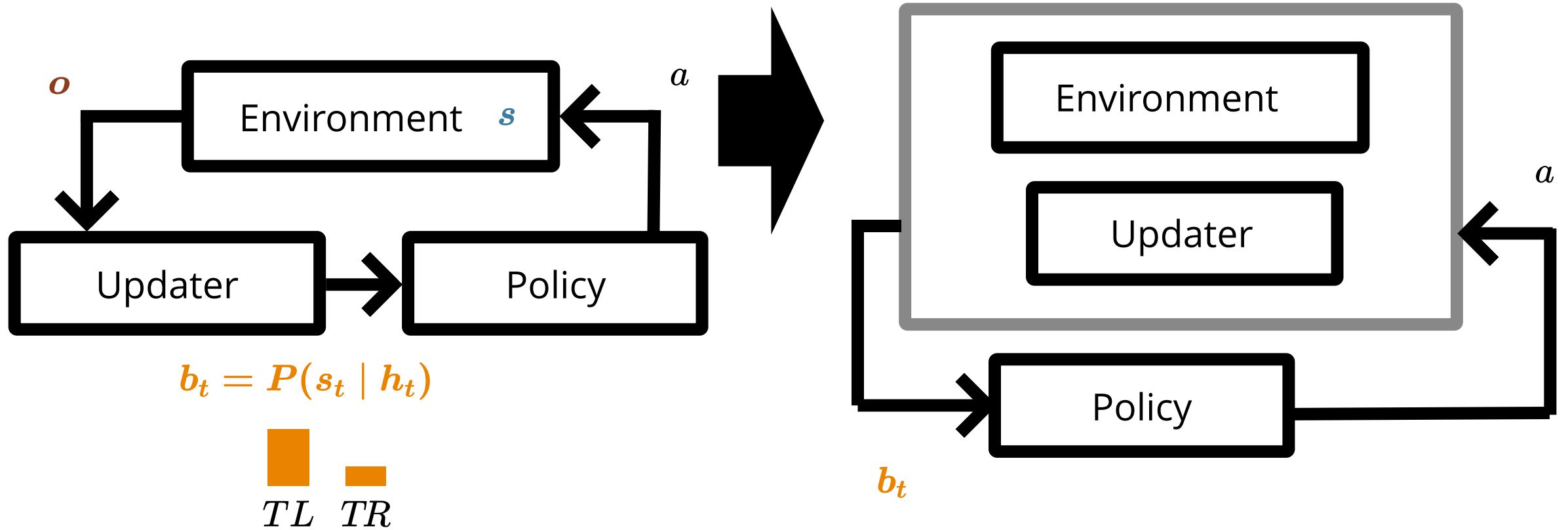
Belief-Space MDP



$$b_t = P(s_t | h_t)$$

TL TR

Belief-Space MDP



Belief Dynamics

Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

Belief Dynamics

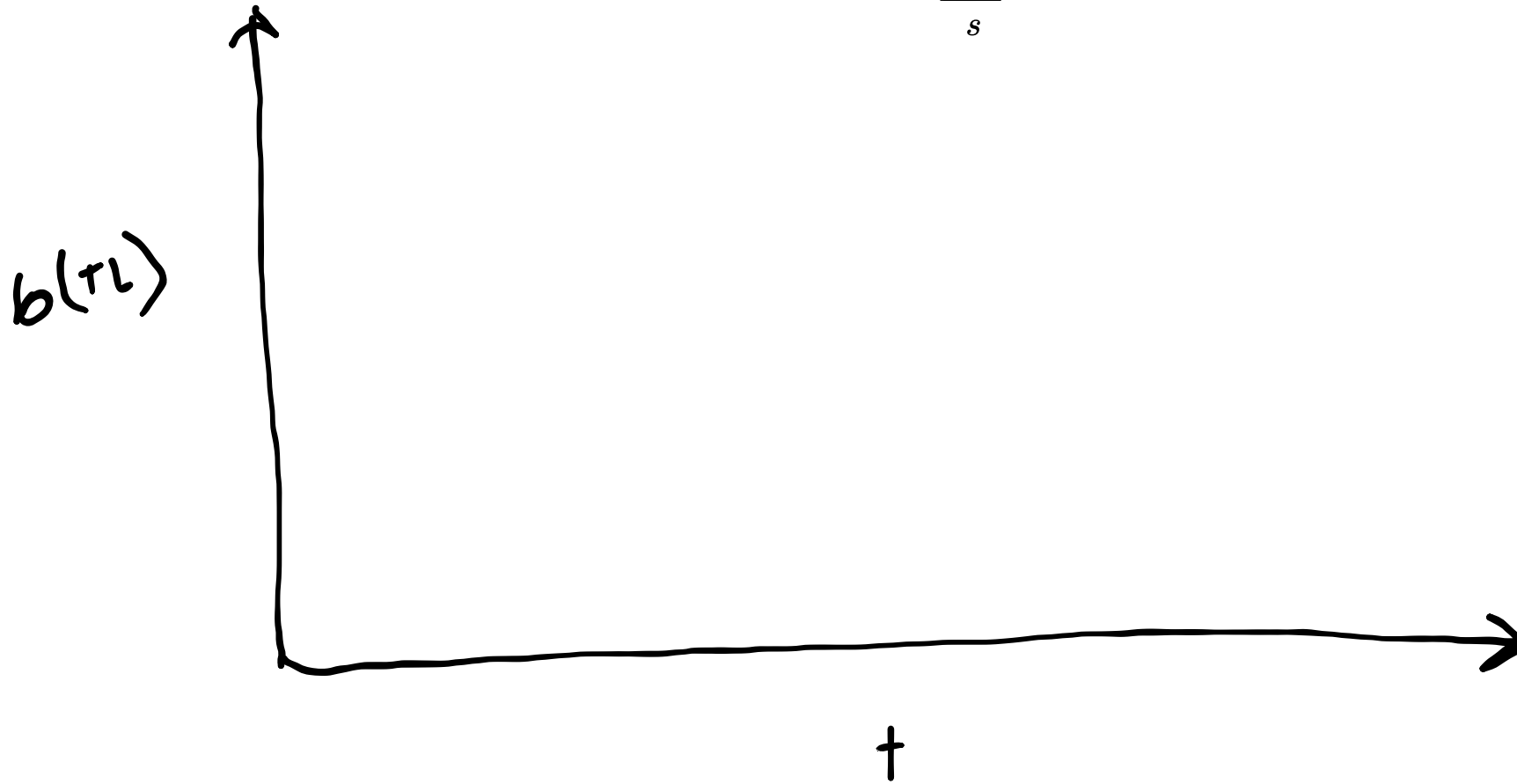
$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$

$b(\tau_L)$

†

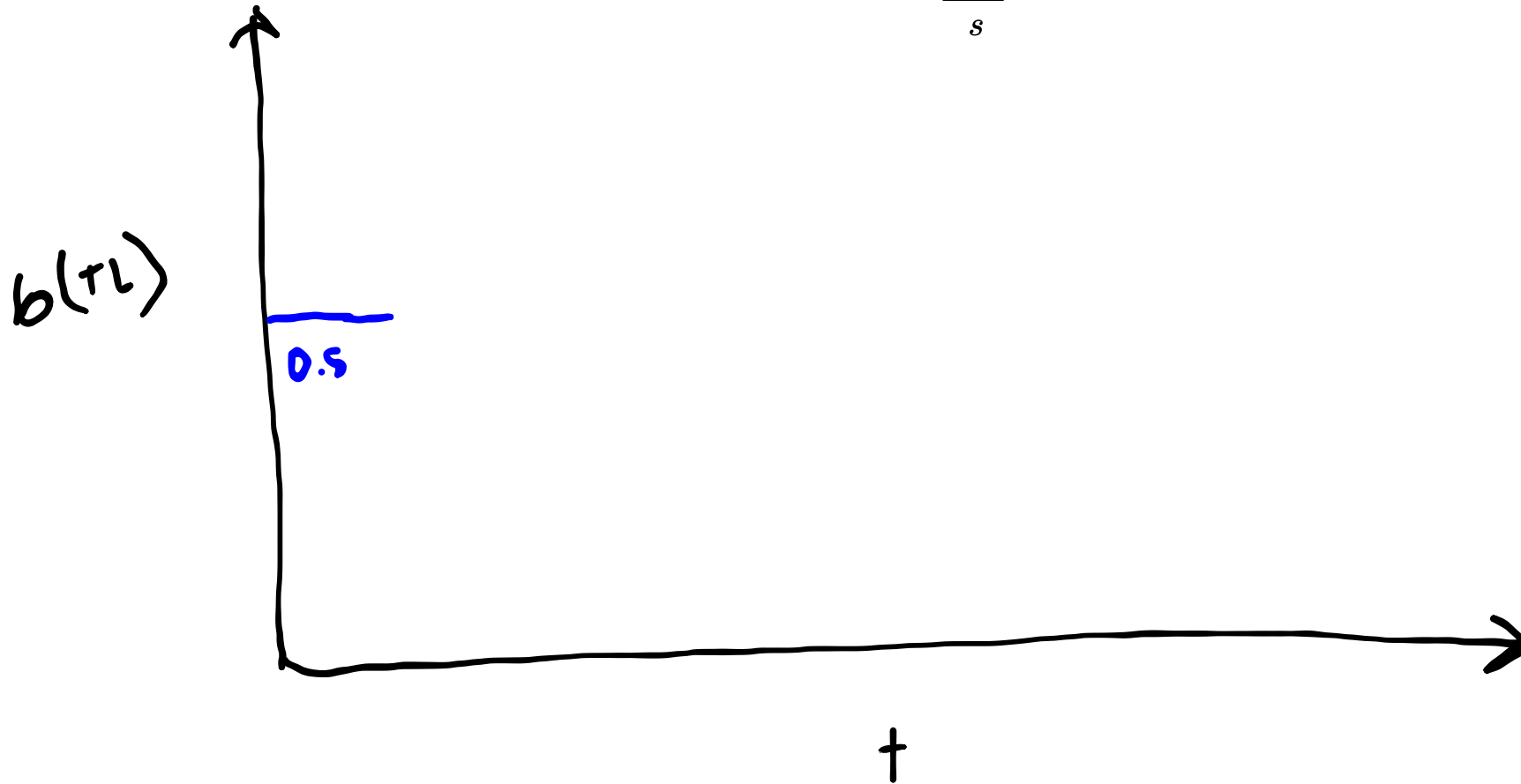
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



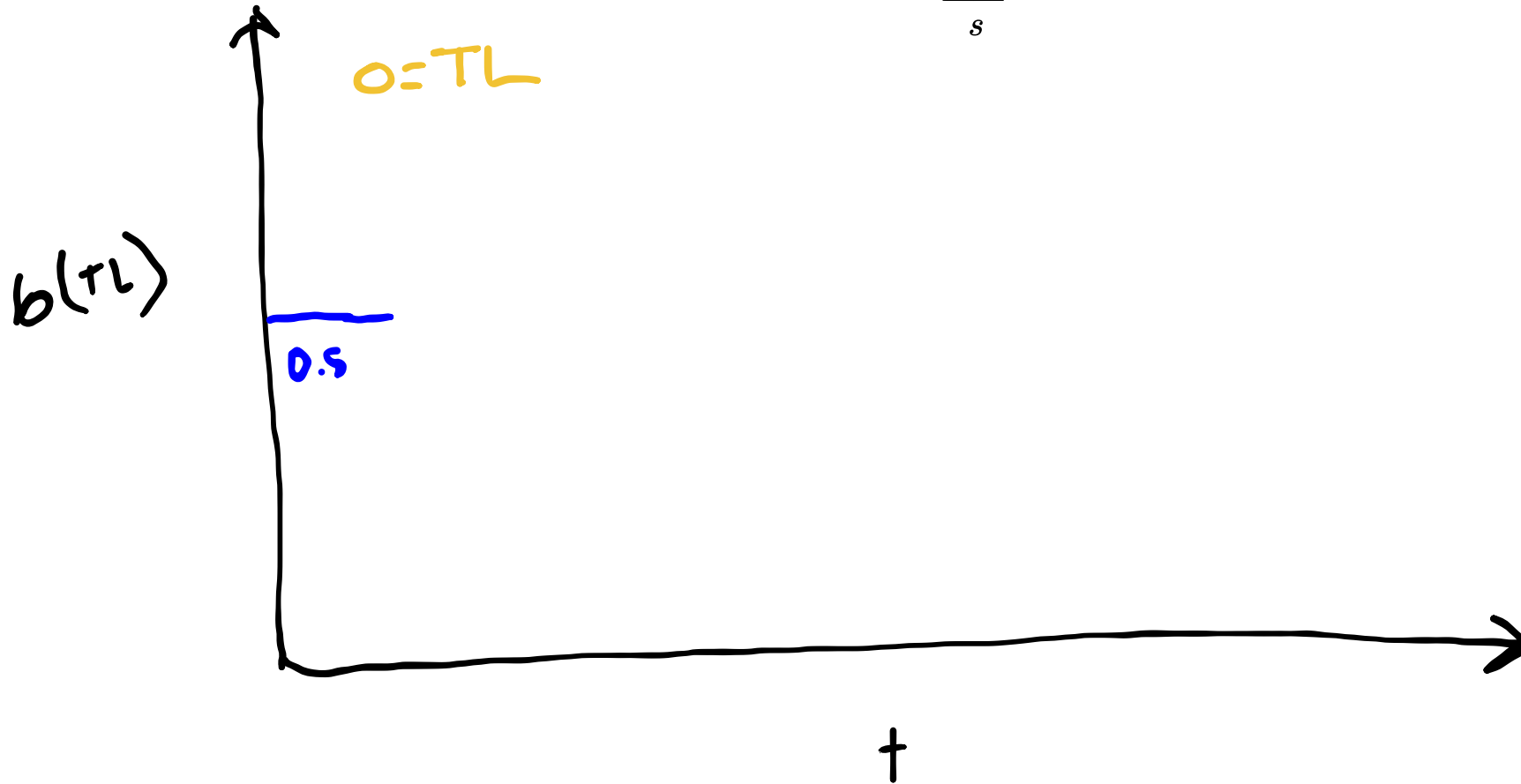
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



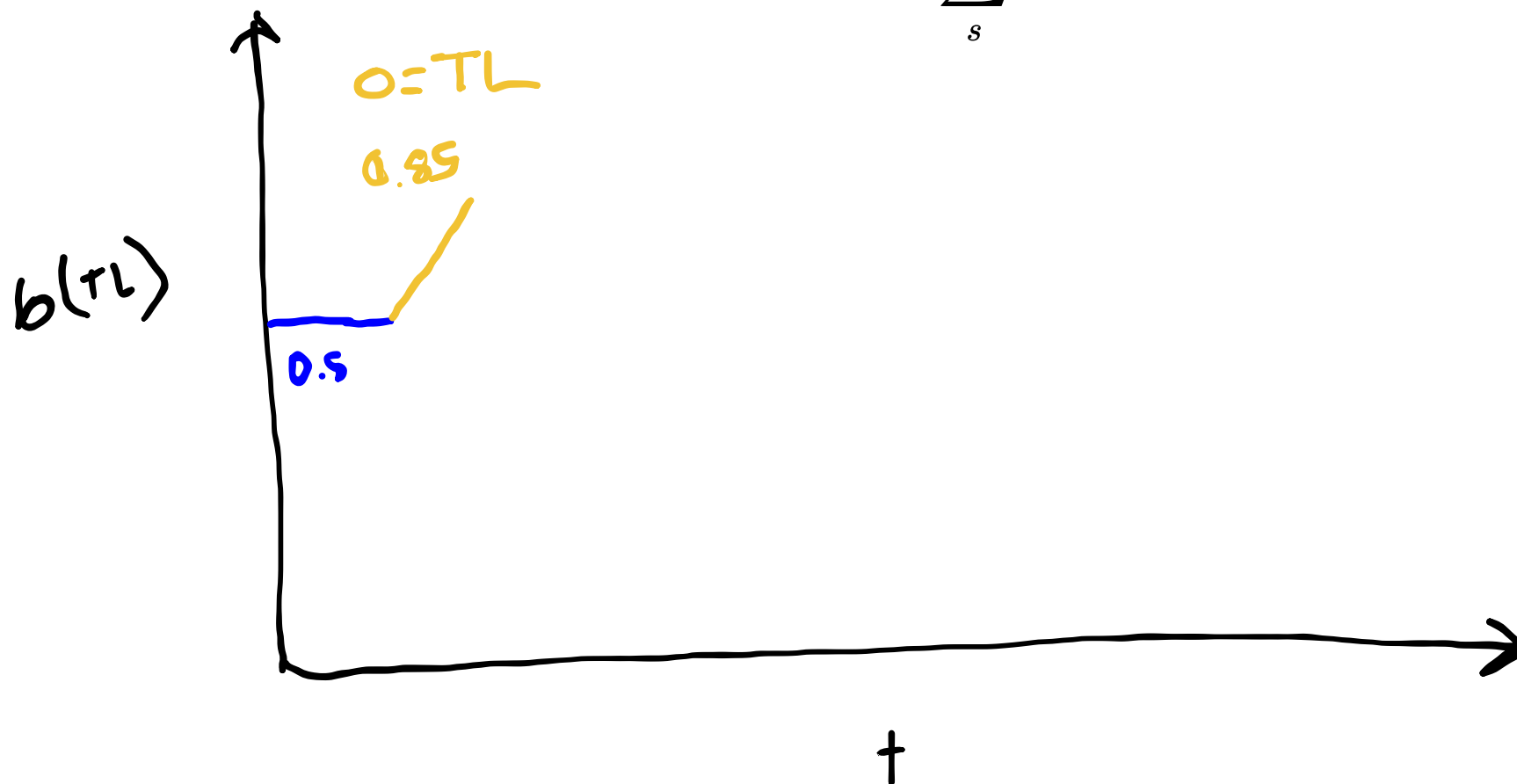
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



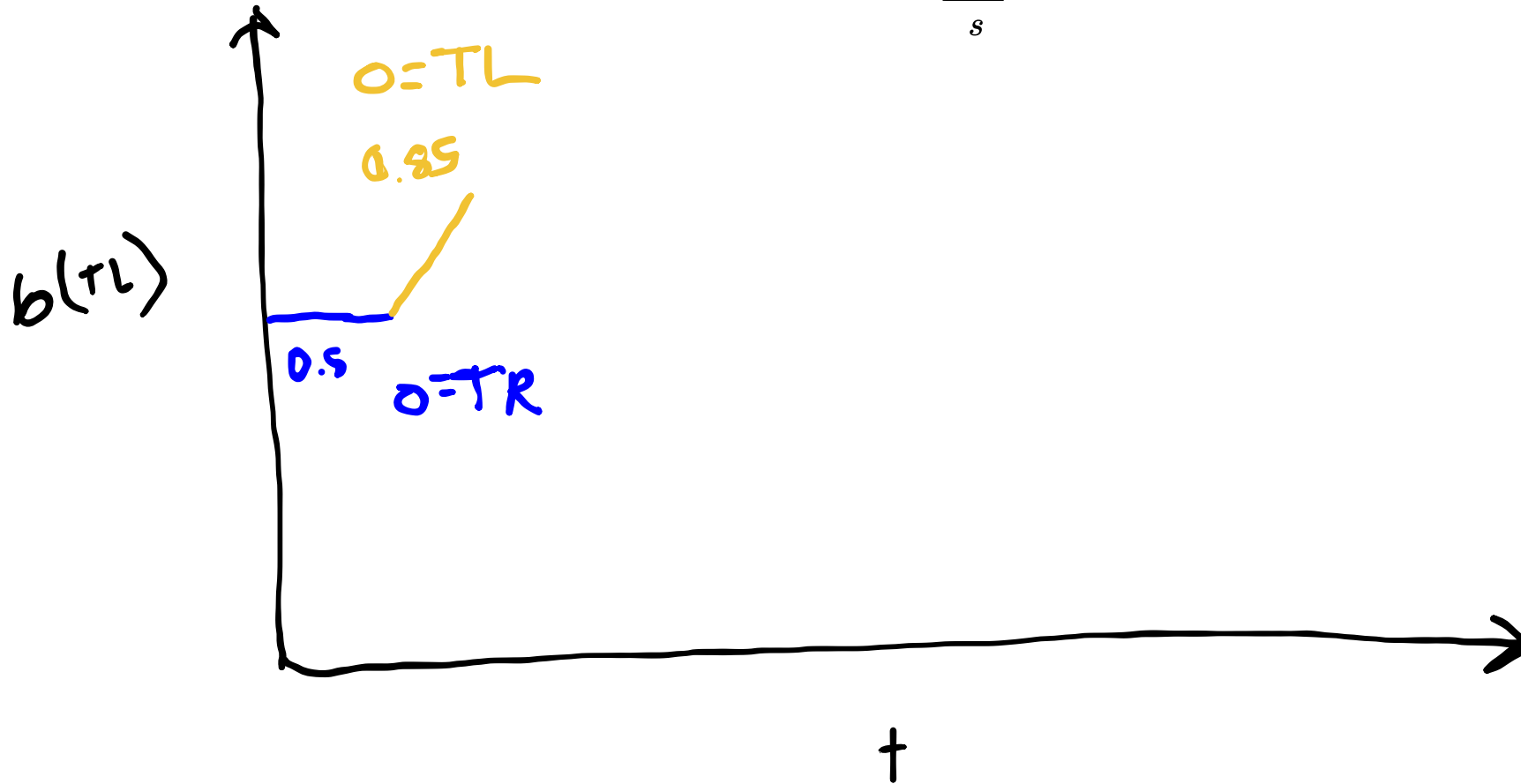
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



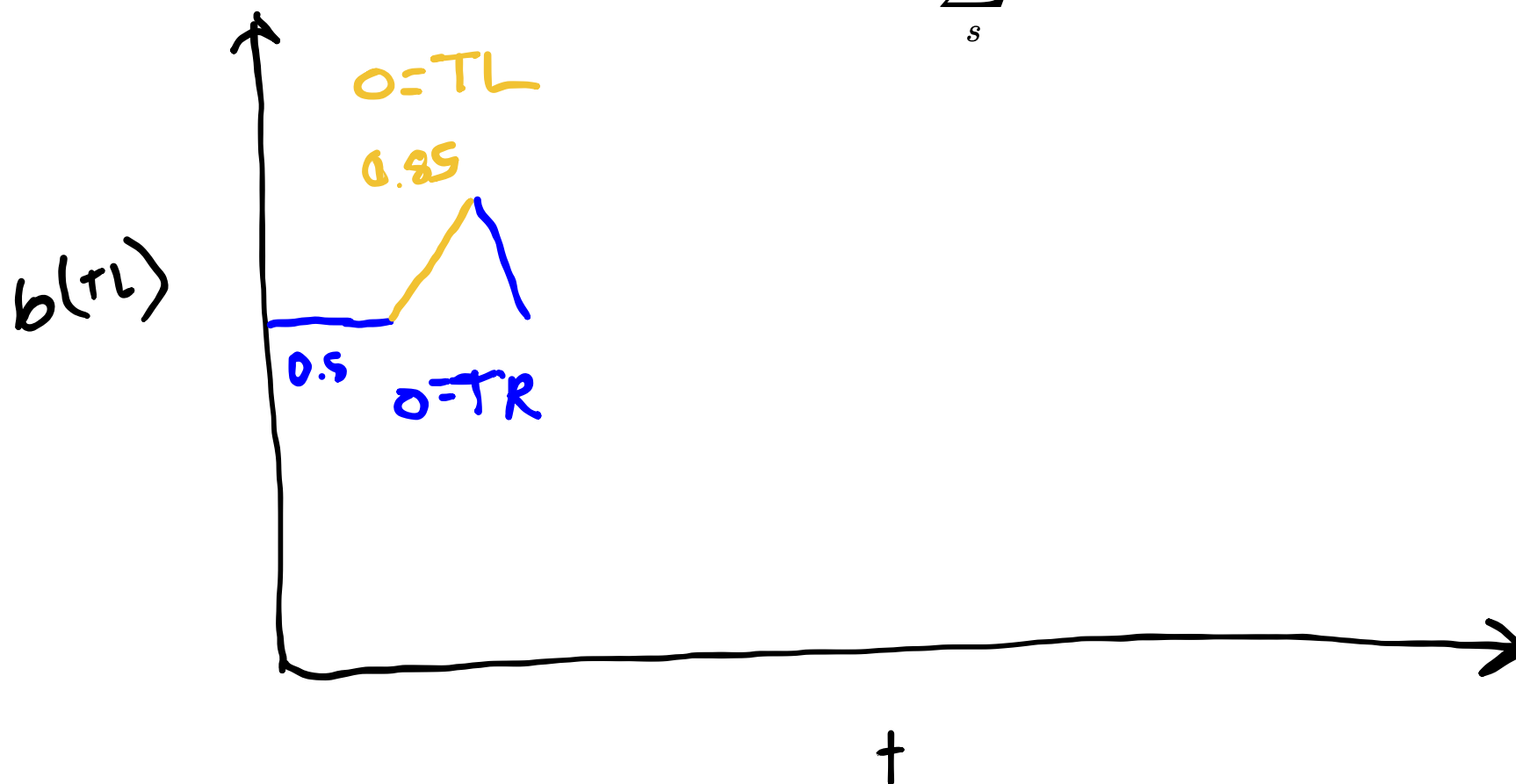
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



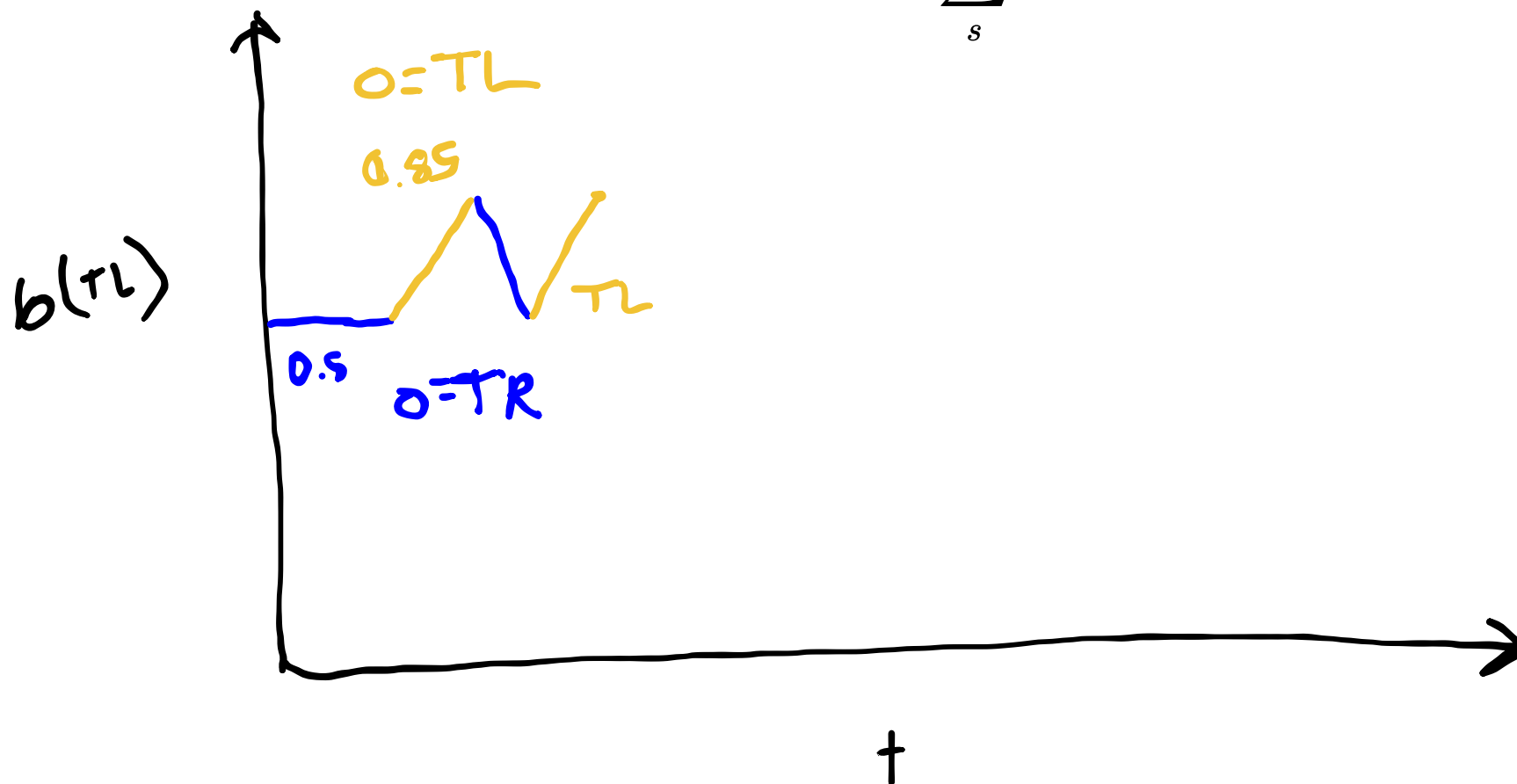
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



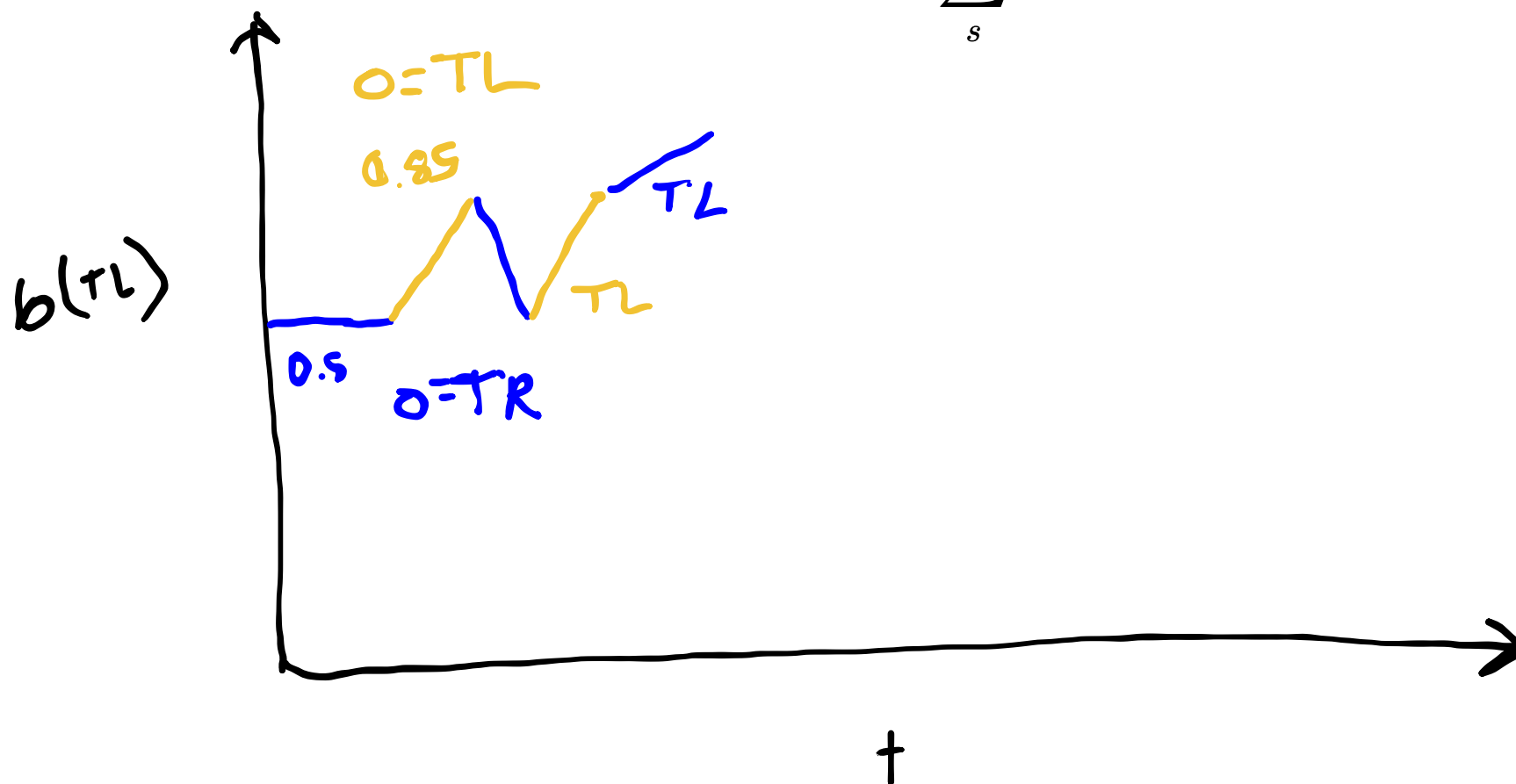
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



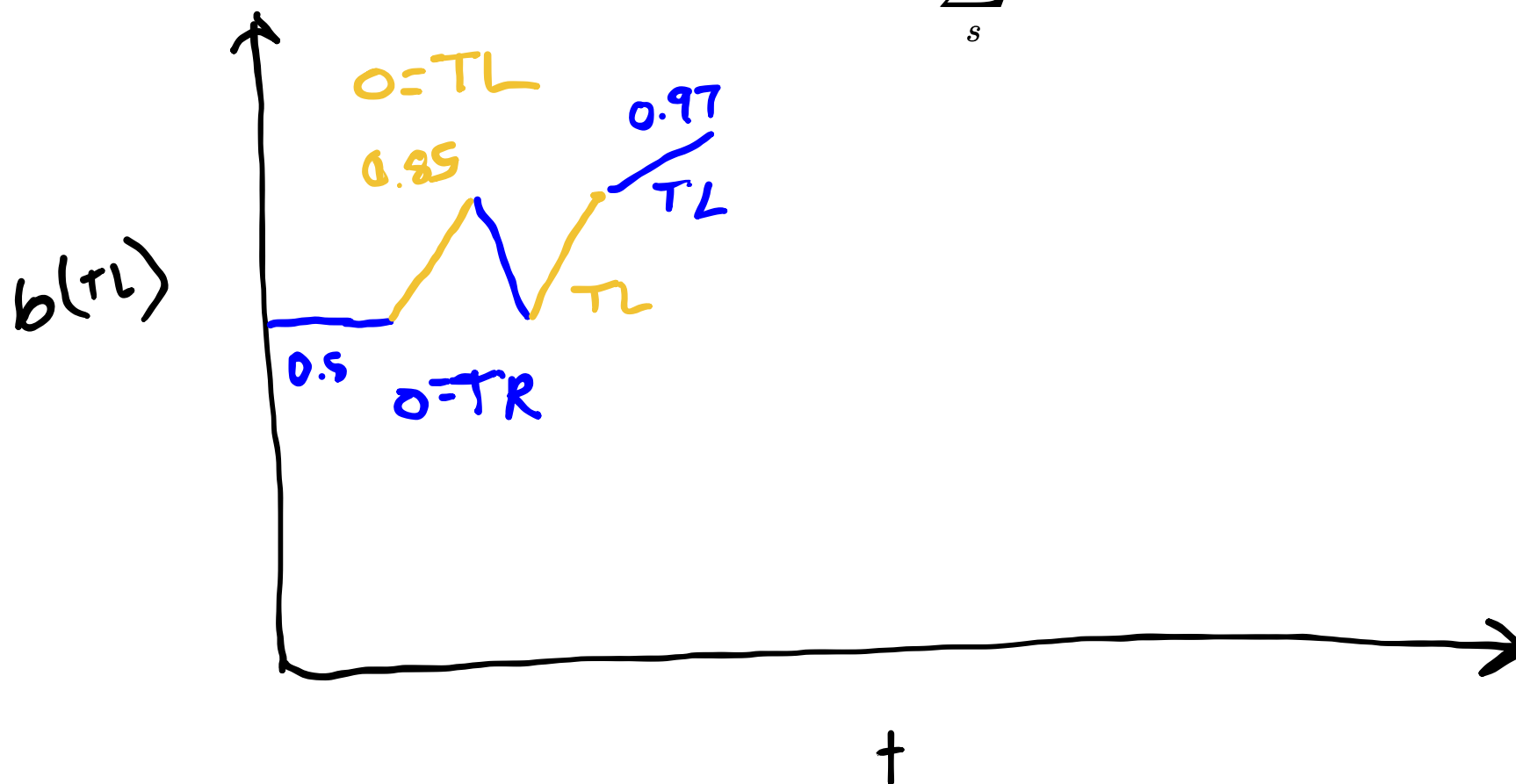
Belief Dynamics

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



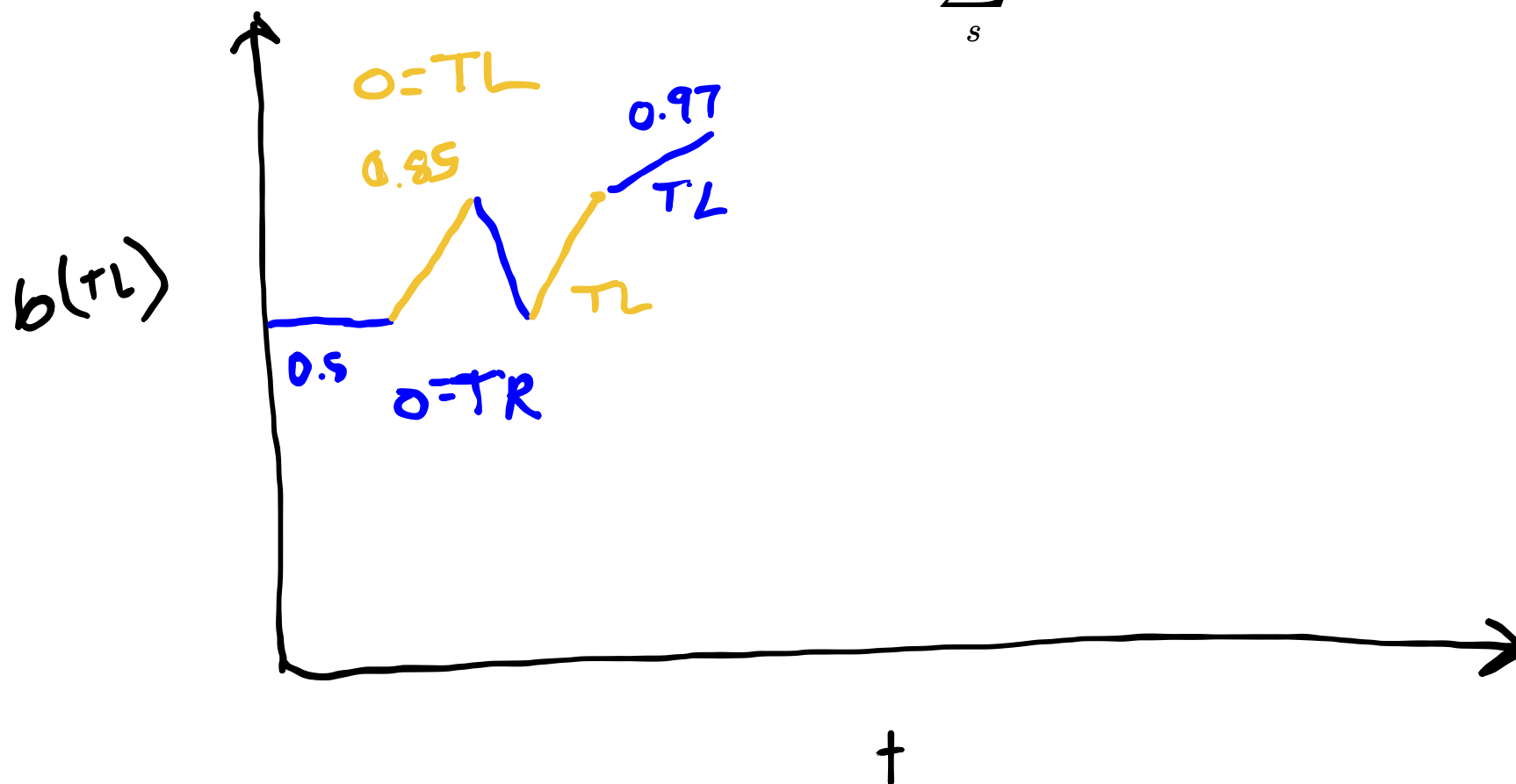
Belief Dynamics

$$b'(s') \propto Z(o | a, s') \sum_s T(s' | s, a) b(s)$$



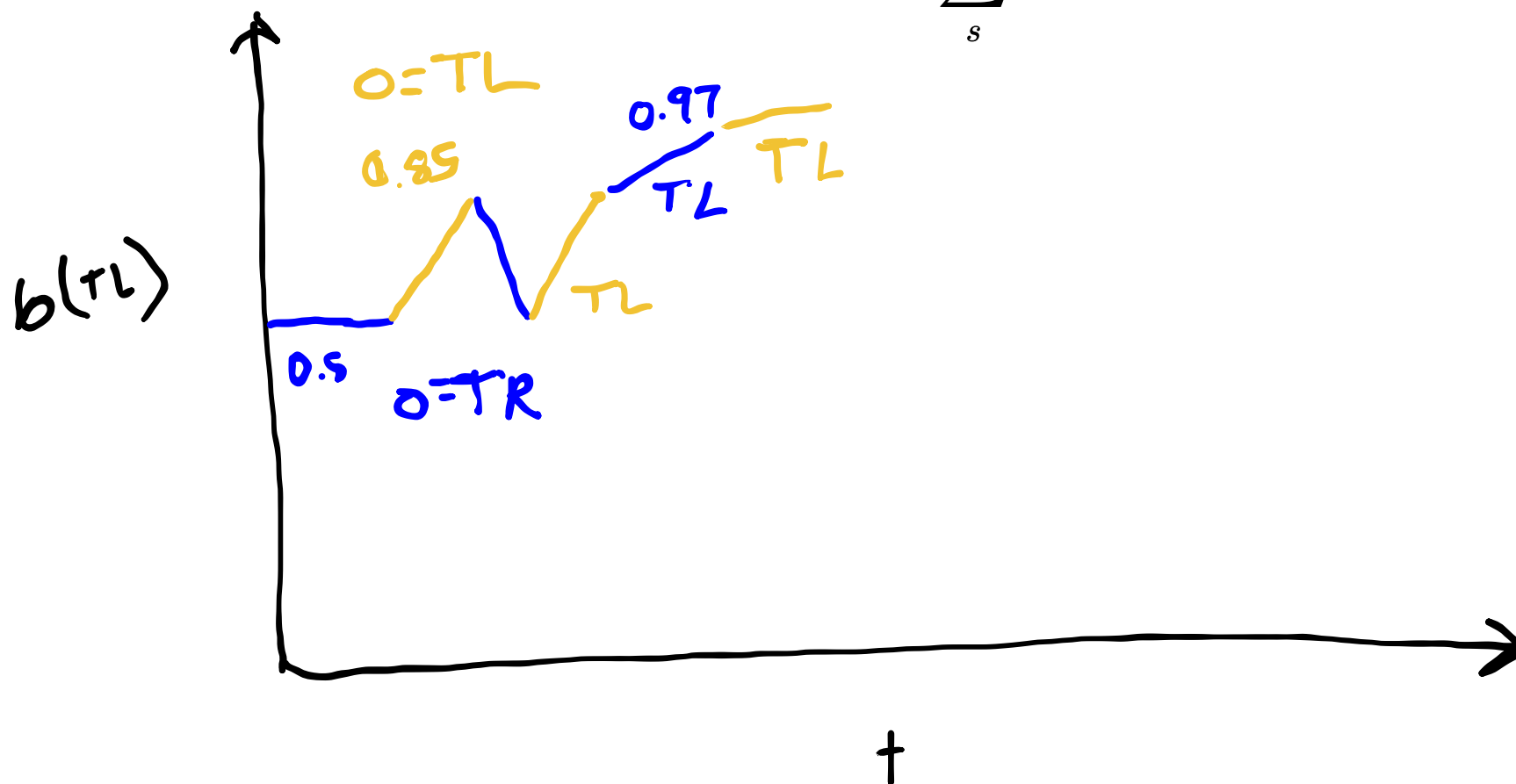
Belief Dynamics

$$b'(s') \propto Z(o | a, s') \sum_s T(s' | s, a) b(s)$$



Belief Dynamics

$$b'(s') \propto Z(o | a, s') \sum_s T(s' | s, a) b(s)$$



Reward

- +10 opening empty
- 1 listening ←
- 100 opening tiger door
- 0 do nothing

One-step utility

(reward)

Want
 $R(b, a)$

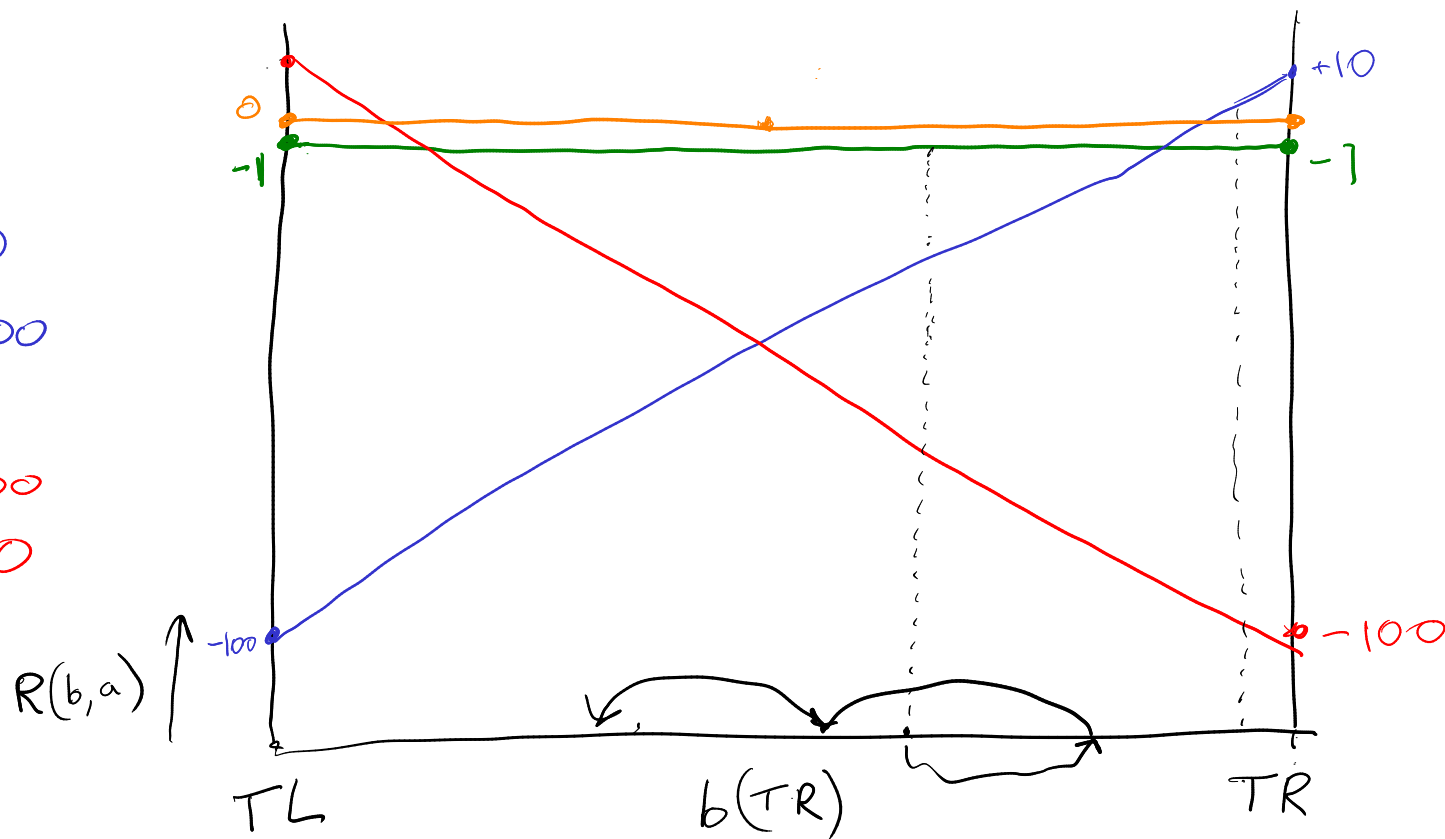
one-step
α-vector
for action α

$$R(b, a) = \vec{R}^a \cdot \vec{b}$$

$\downarrow$
 $\begin{bmatrix} R(s^1, a) \\ R(s^2, a) \\ \vdots \\ R(s^n, a) \end{bmatrix}$

$\downarrow$
 $\begin{bmatrix} b(s^1) \\ b(s^2) \\ \vdots \\ b(s^n) \end{bmatrix}$

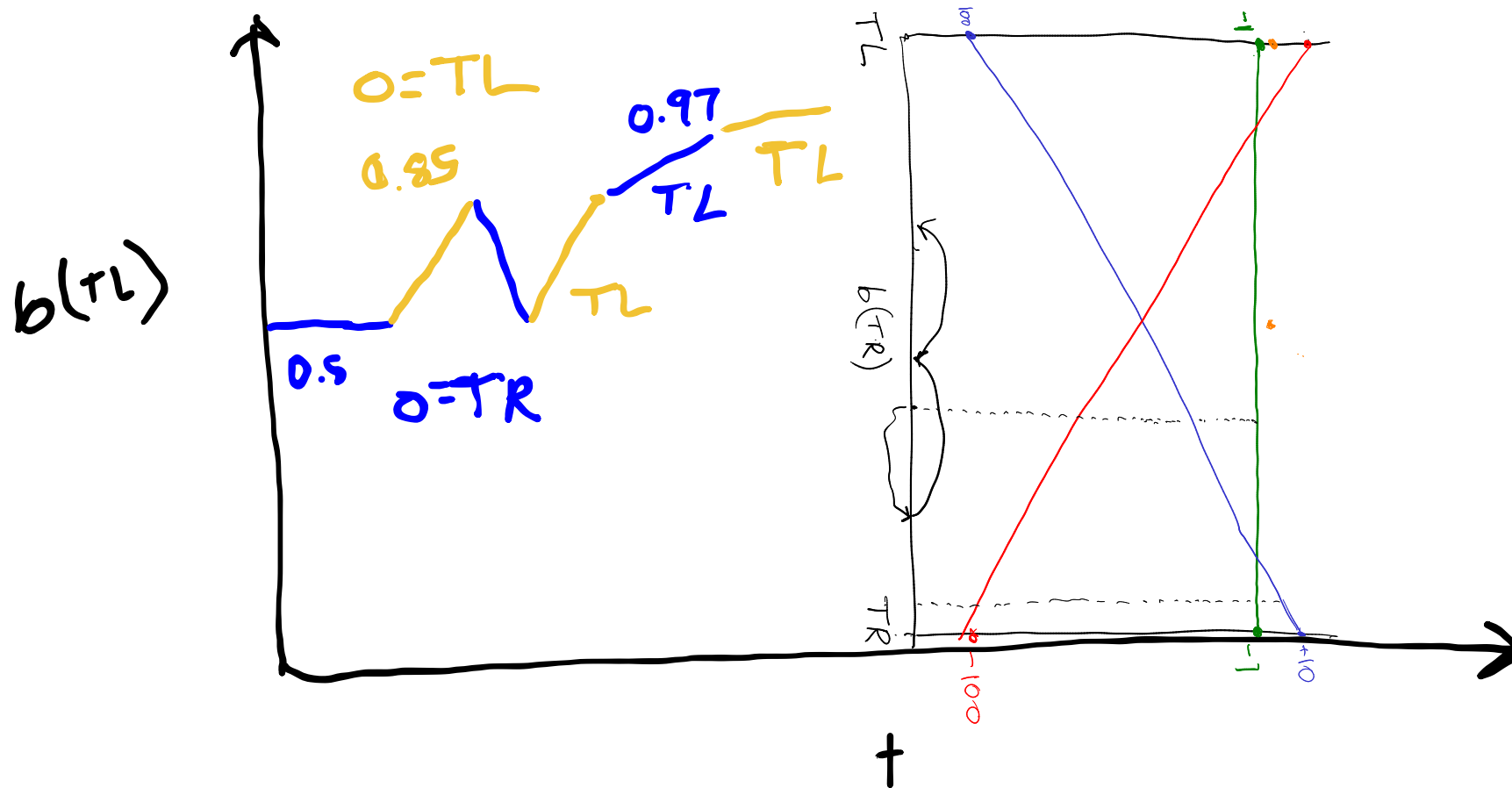
$$\pi_{\text{greedy}}(b) = \arg\max_a \vec{R}^a \cdot \vec{b}$$



$$R(b, a) = E[R(s, a)] = \sum_s b(s) R(s, a) = b(TL) R(TL, a) + (1 - b(TL)) R(TR, a)$$

α -vectors define a policy

$$b'(s') \propto Z(o \mid a, s') \sum_s T(s' \mid s, a) b(s)$$



Exercise 2: Crying Baby 1-Step Utility

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

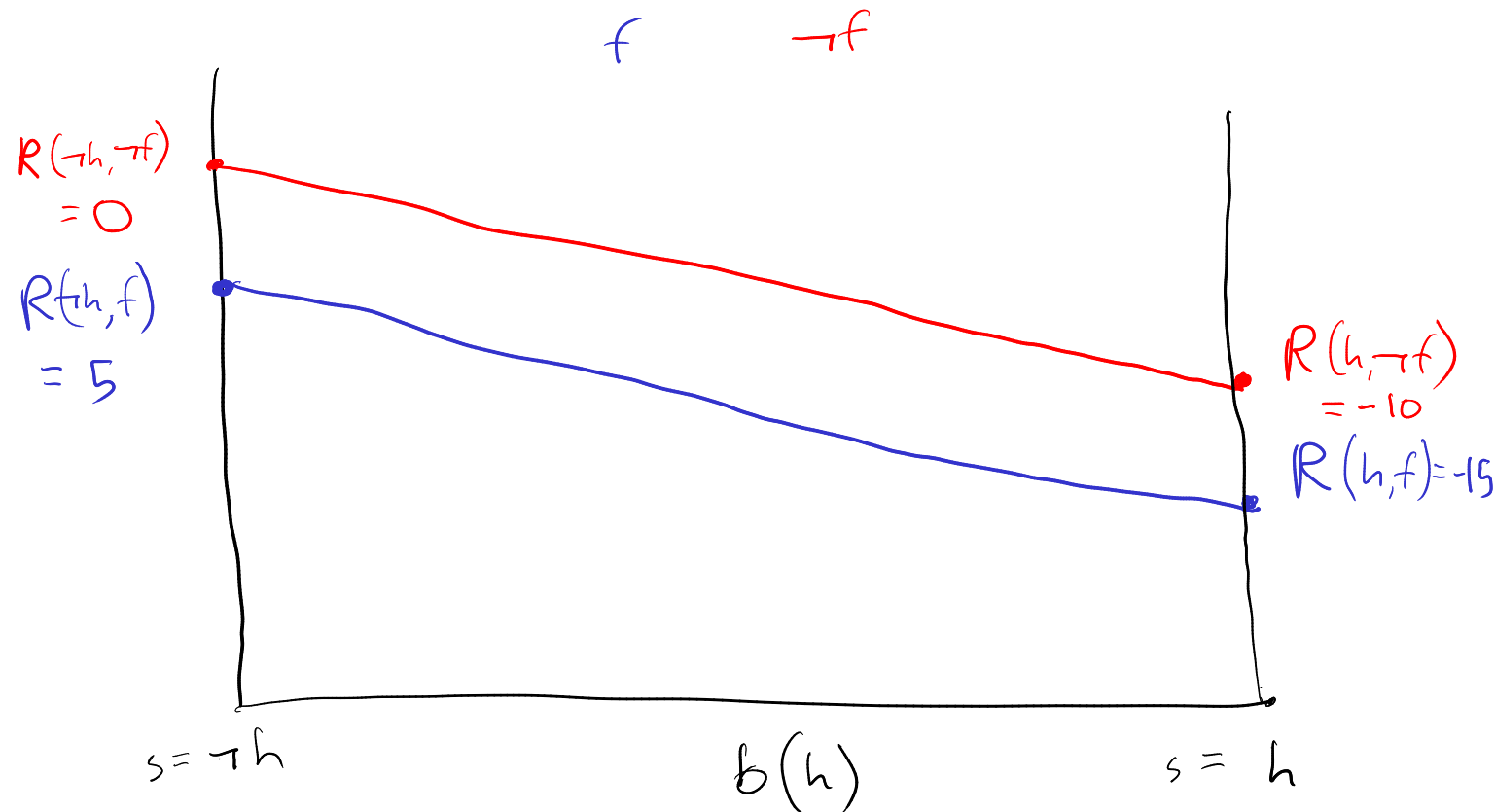
$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

$$\gamma = 0.9$$

Draw the 1-step utility α -vectors for the Crying Baby problem.



Alpha Vectors for Conditional Plans

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (oL) (oR)

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

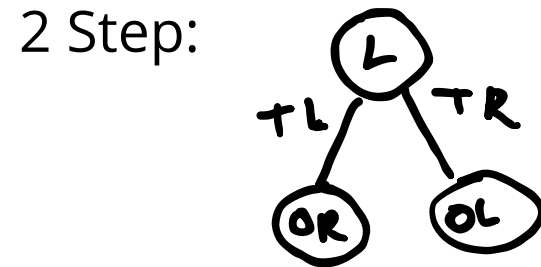
1 Step: (L) (oL) (oR)

2 Step:

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

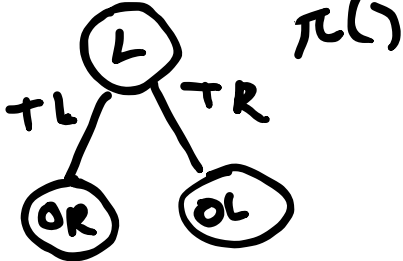
1 Step: (L) (OL) (OR)



Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

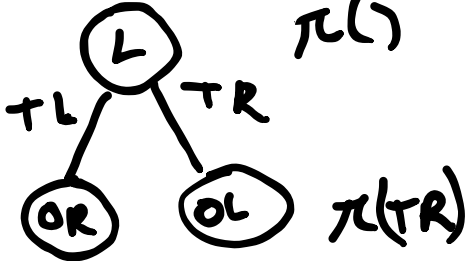
1 Step: (L) (OL) (OR)

2 Step: 

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

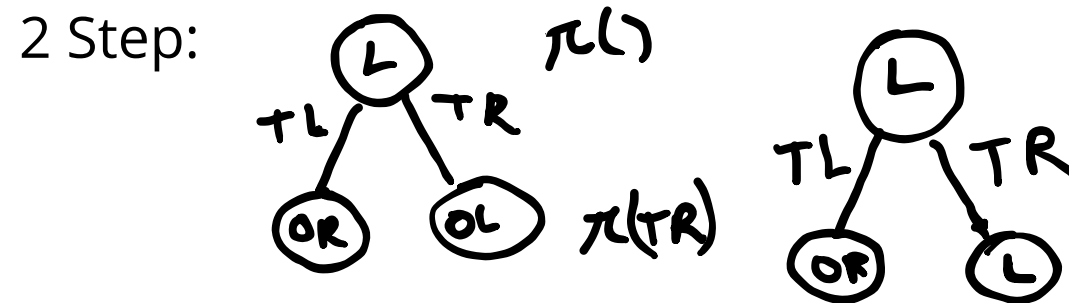
1 Step: (L) (OL) (OR)

2 Step: 

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

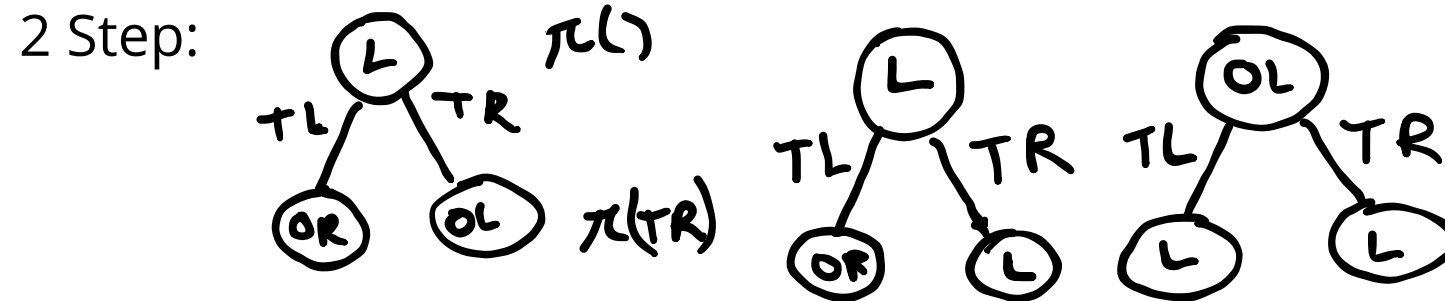
1 Step: (L) (OL) (OR)



Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)

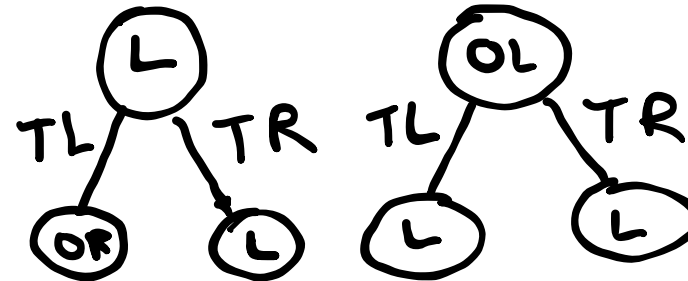


Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)

2 Step: (L) $\pi(L)$
TL TR
(OR) (OL) $\pi(TR)$

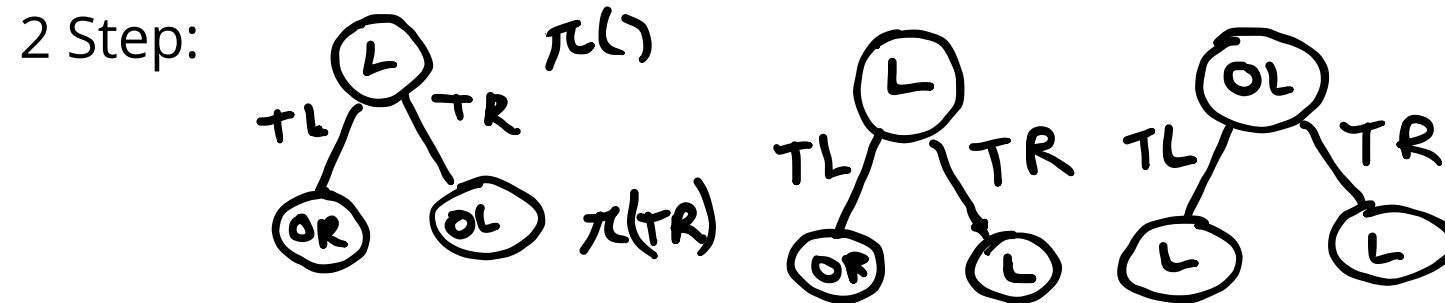


$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

Alpha Vectors for Conditional Plans

Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)



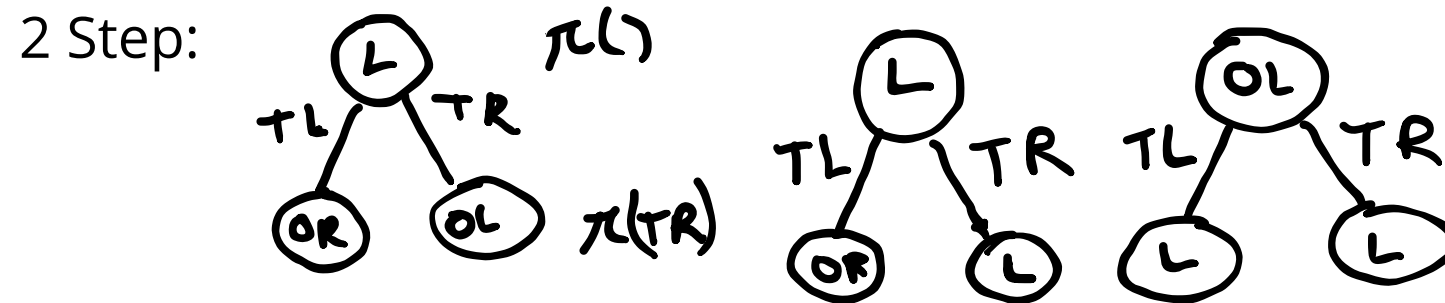
$$|A| \frac{(|O|^h - 1)}{(|O| - 1)}$$

27 two step plans!

Alpha Vectors for Conditional Plans

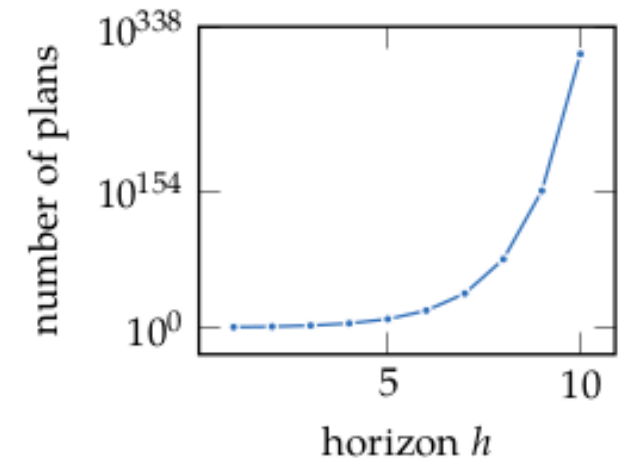
Conditional Plans: fixed-depth history-based policies

1 Step: (L) (OL) (OR)



$$|A|^{\frac{(|O|^h - 1)}{(|O| - 1)}}$$

27 two step plans!

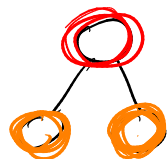


Alpha Vectors for Conditional Plans

Alpha Vectors for Conditional Plans

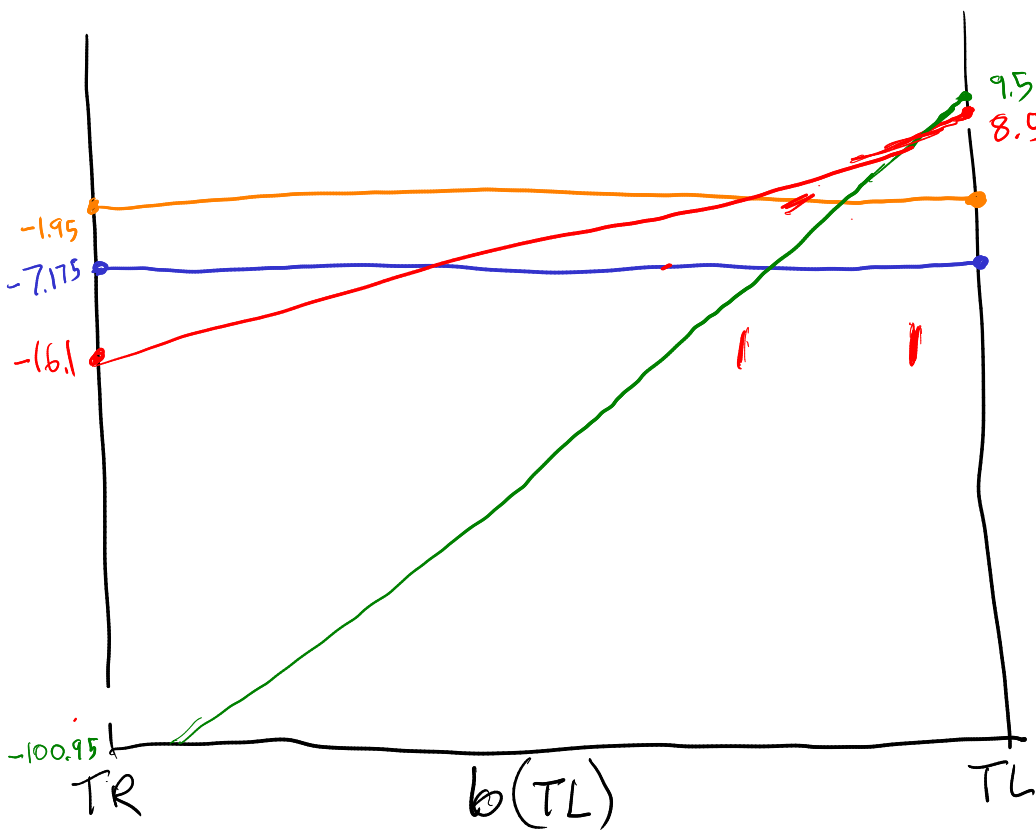
For 1-step: $U^\pi(s) = R(s, \pi())$

Alpha Vectors for Conditional Plans



For 1-step: $U^\pi(s) = R(s, \pi())$

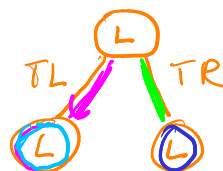
$$U^\pi(s) = \underline{R(s, \pi())} + \gamma \left[\sum_{s'} T(s' | s, \pi()) \sum_o \cancel{\pi(o | \pi(), s')} \underline{U^{\pi(o)}(s')} \right]$$



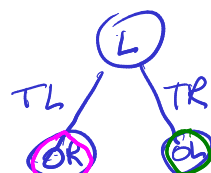
①
 $U^L(\cdot) = -1$

②
 $U^{OL}(TR) = 10$
 $U^{OL}(TL) = -100$

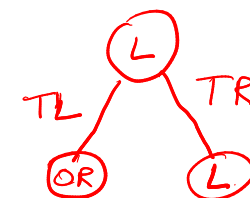
③
 $U^{OR}(TR) = -100$
 $U^{OR}(TL) = 10$



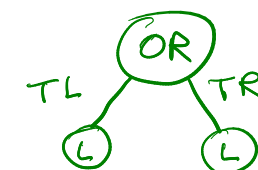
$U^\pi(TL) = -1 + \gamma [1(0.85(-1) + 0.15(-1))]$
 $= -1 + 0.95(-1) = -1.95$
 $U^\pi(TR) = -1.95$



$U^\pi(TL) = -1 + \gamma [1(0.85 \cdot 10 + 0.15(-100))]$
 $= -7.175$
 $U^\pi(TR) = -7.175$



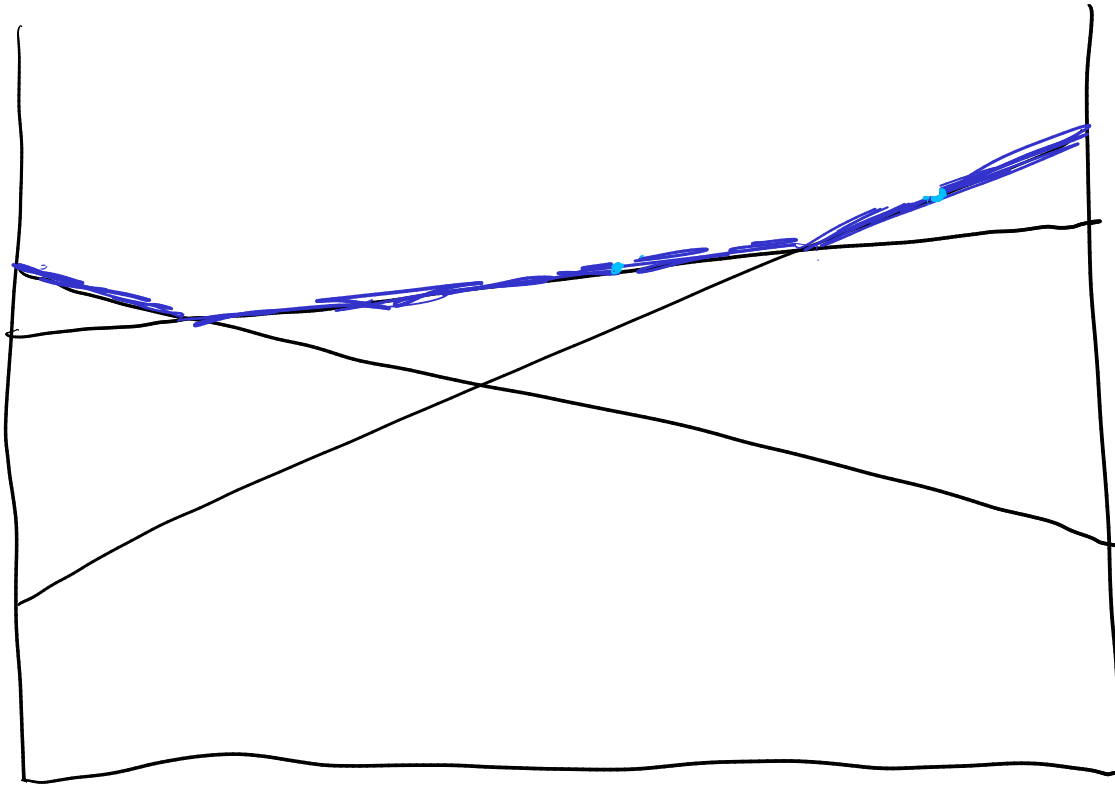
$U^\pi(TL) = 8.5$
 $U^\pi(TR) = -16$
 $-1 + \gamma(0.15(100) + 0.85(-1)) = -16.1$



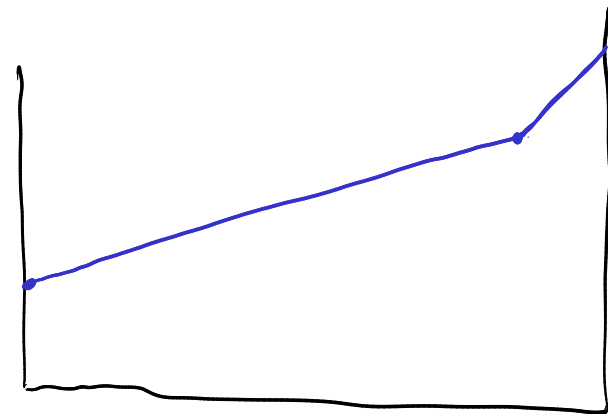
$U^\pi(TL) = 9.5$
 $U^\pi(TR) = -100.95$

POMDP Value Functions

POMDP Value Functions



$$\underline{V^*(b)} = \max_{\alpha \in \Gamma} \alpha^\top b$$



Exercise: 2-Step Crying Baby α Vectors

$$S = \{h, \neg h\} \quad T(h \mid h, \neg f) = 1.0$$

$$A = \{f, \neg f\} \quad T(h \mid \neg h, \neg f) = 0.1$$

$$O = \{c, \neg c\} \quad T(\neg h \mid \cdot, f) = 1.0$$

$$R(s, a) = R(s) + R(a)$$

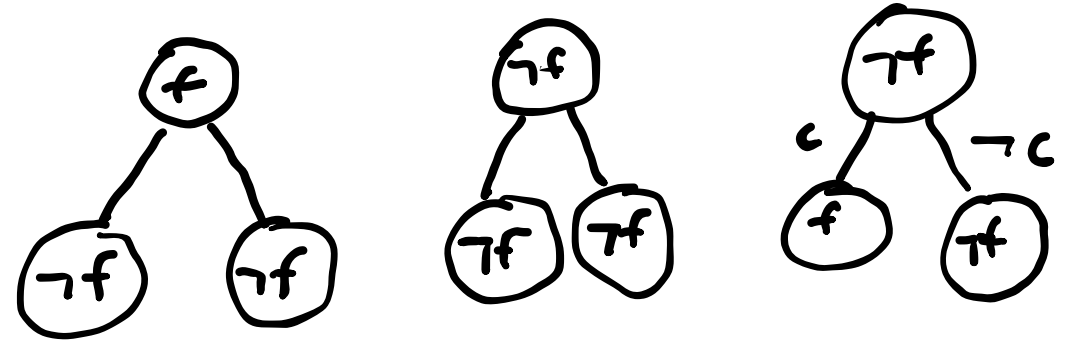
$$R(s) = \begin{cases} -10 & \text{if } s = h \\ 0 & \text{otherwise} \end{cases}$$

$$R(a) = \begin{cases} -5 & \text{if } a = f \\ 0 & \text{otherwise} \end{cases}$$

$$Z(c \mid \cdot, h) = 0.8$$

$$Z(c \mid \cdot, \neg h) = 0.1$$

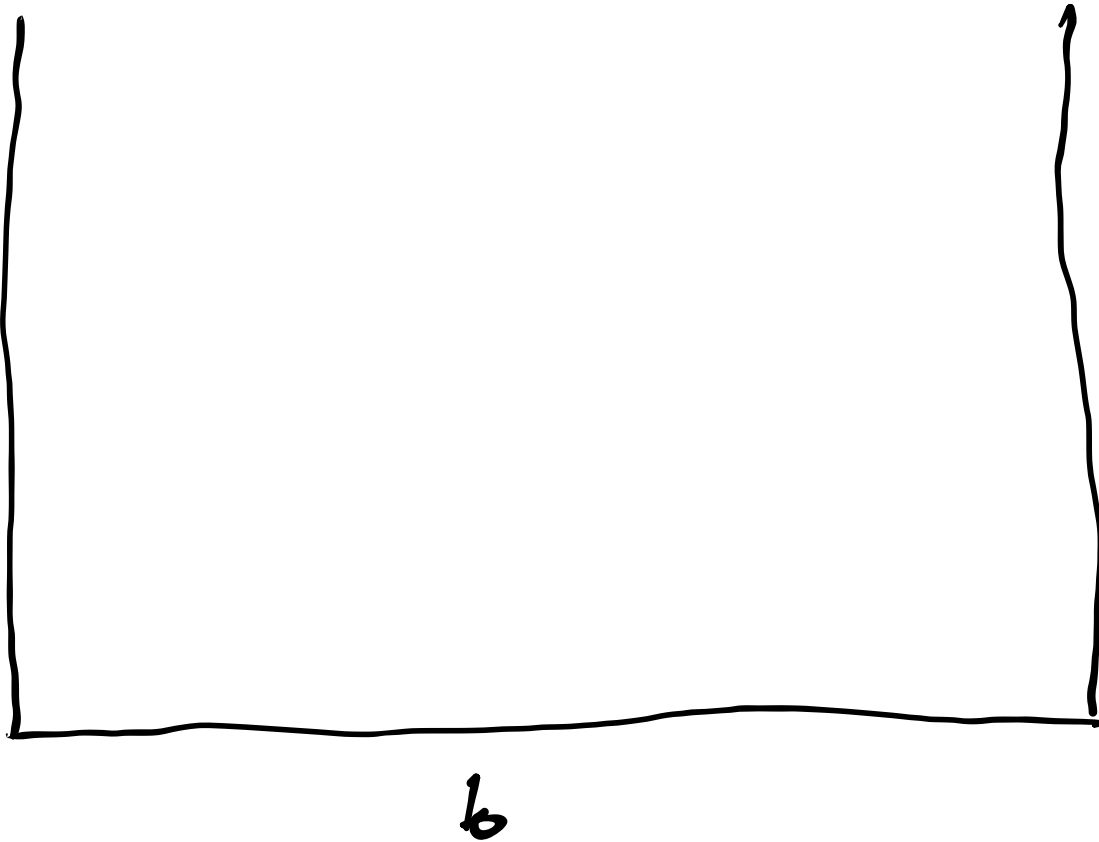
$$\gamma = 0.9$$



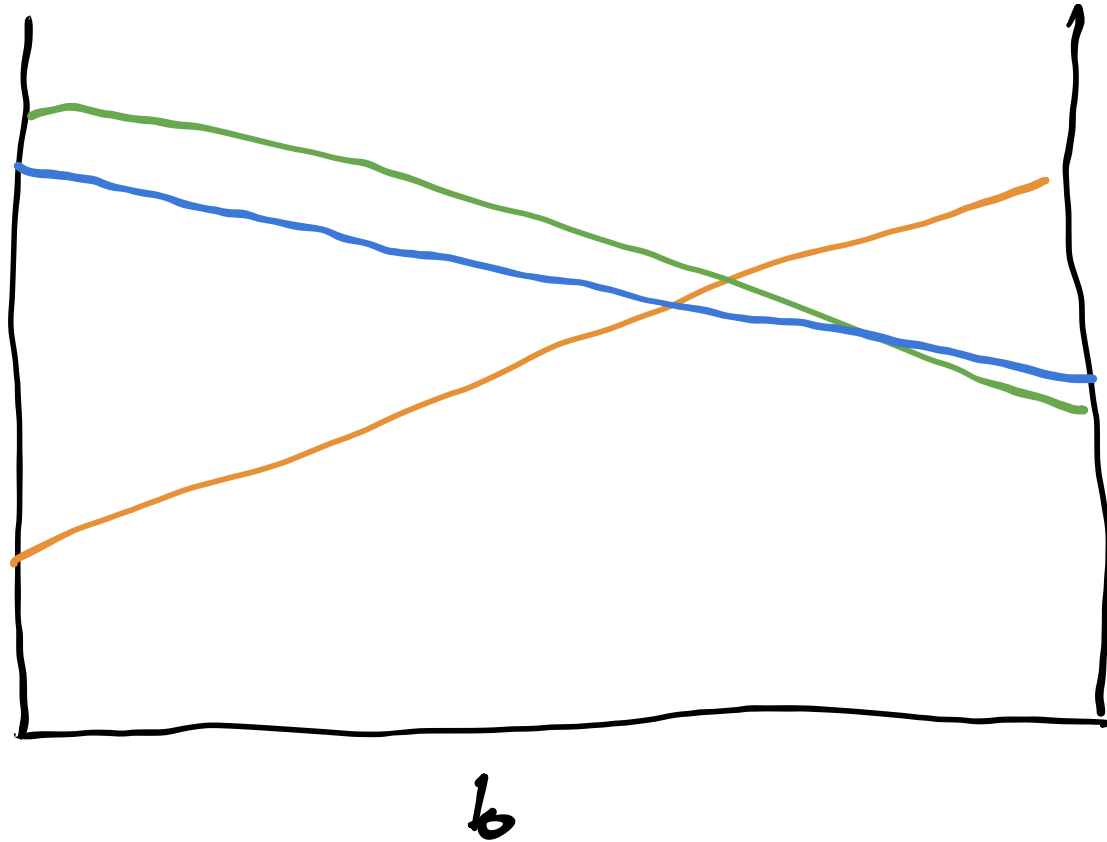
$$U^\pi(s) = R(s, \pi()) + \gamma \left[\sum_{s'} T(s' \mid s, \pi()) \sum_o O(o \mid \pi(), s') U^{\pi(o)}(s') \right]$$

α -Vector Pruning

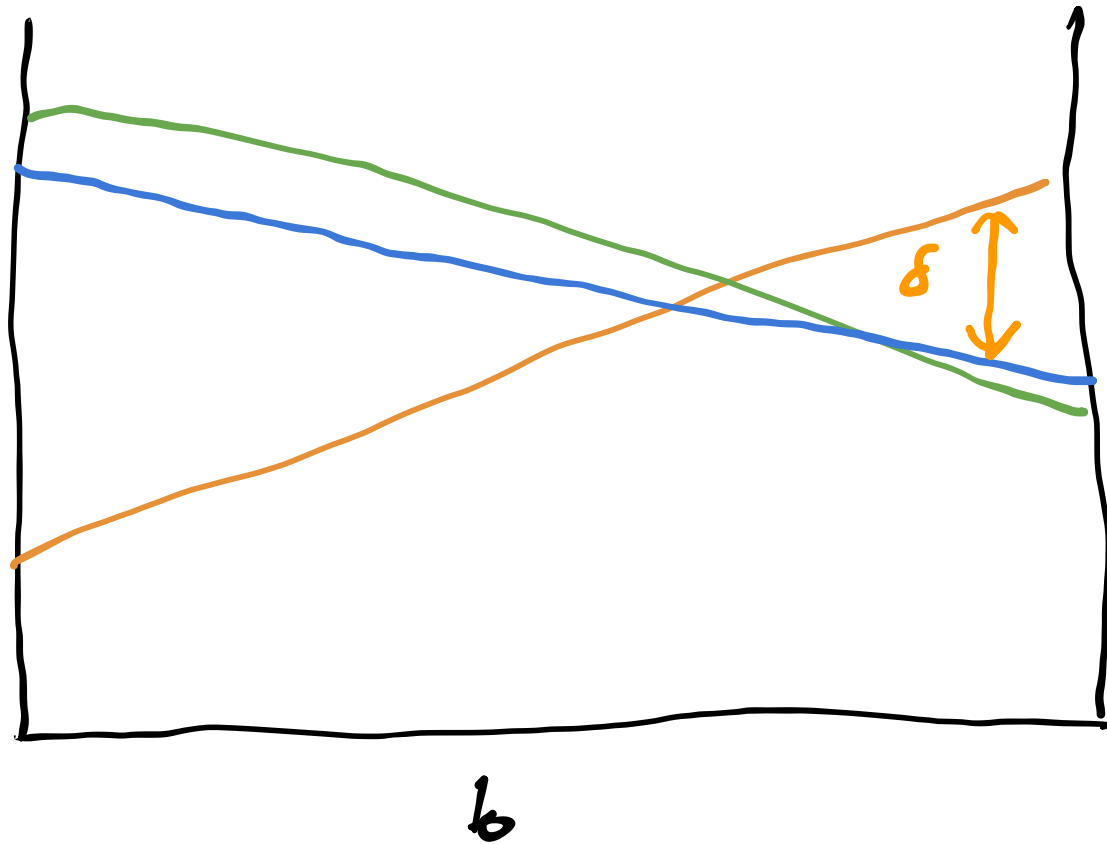
α -Vector Pruning



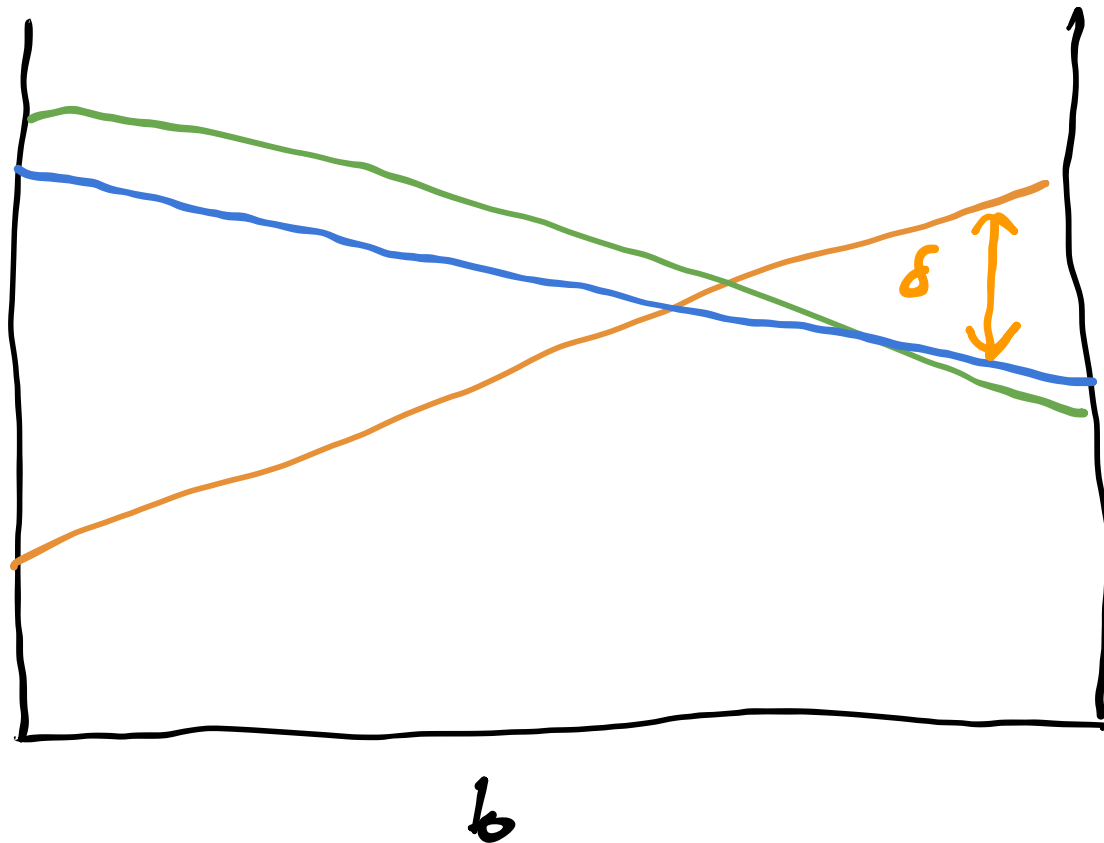
α -Vector Pruning



α -Vector Pruning

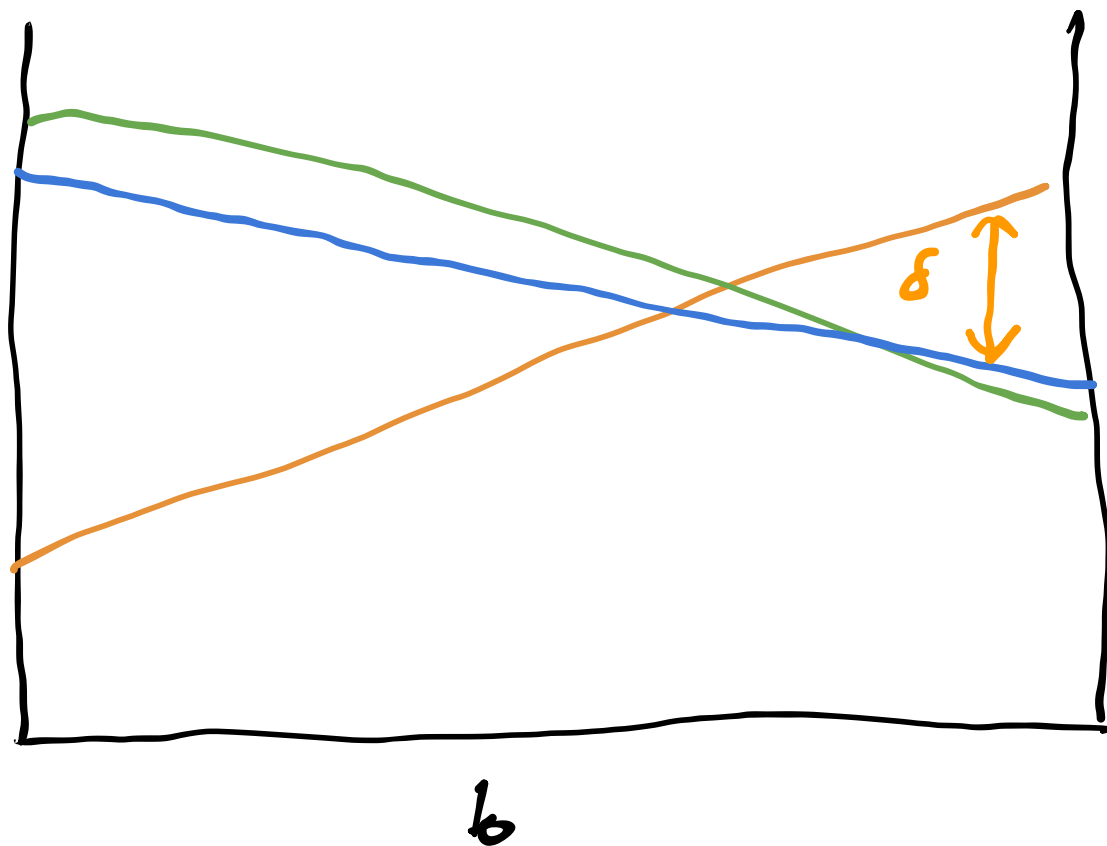


α -Vector Pruning



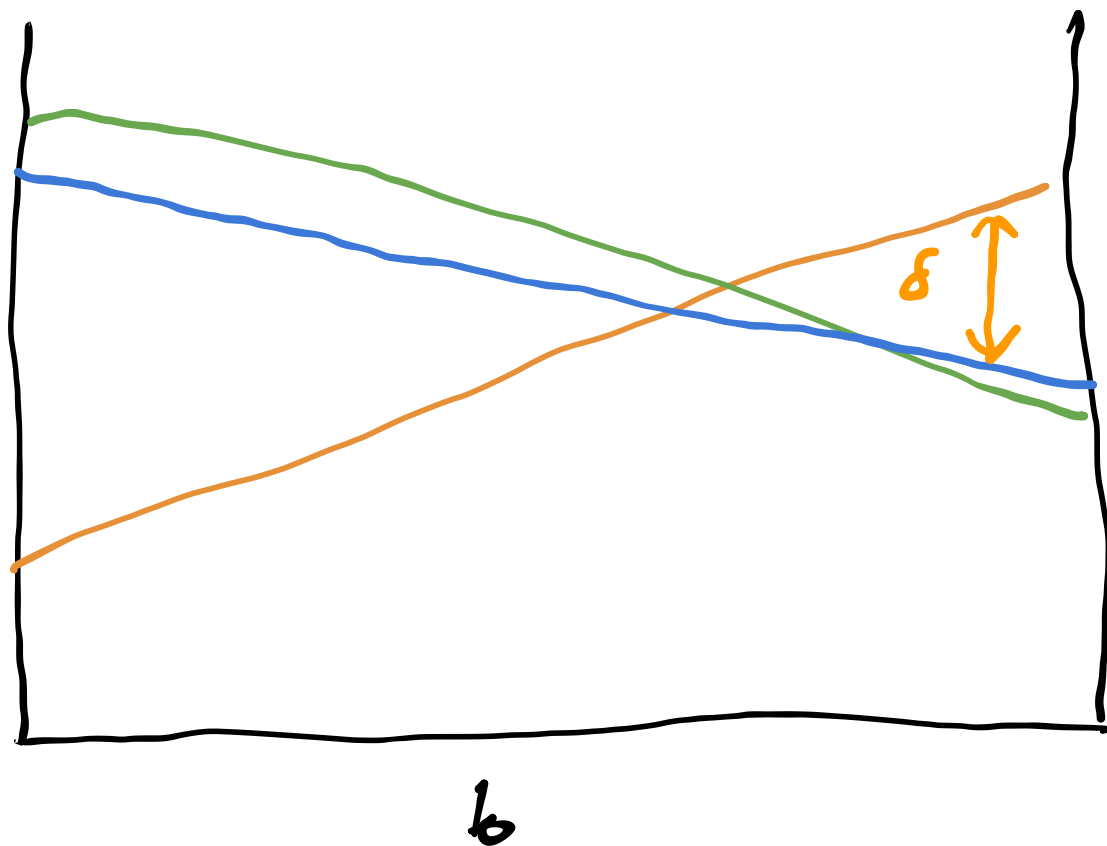
maximize δ
 δ, b

α -Vector Pruning



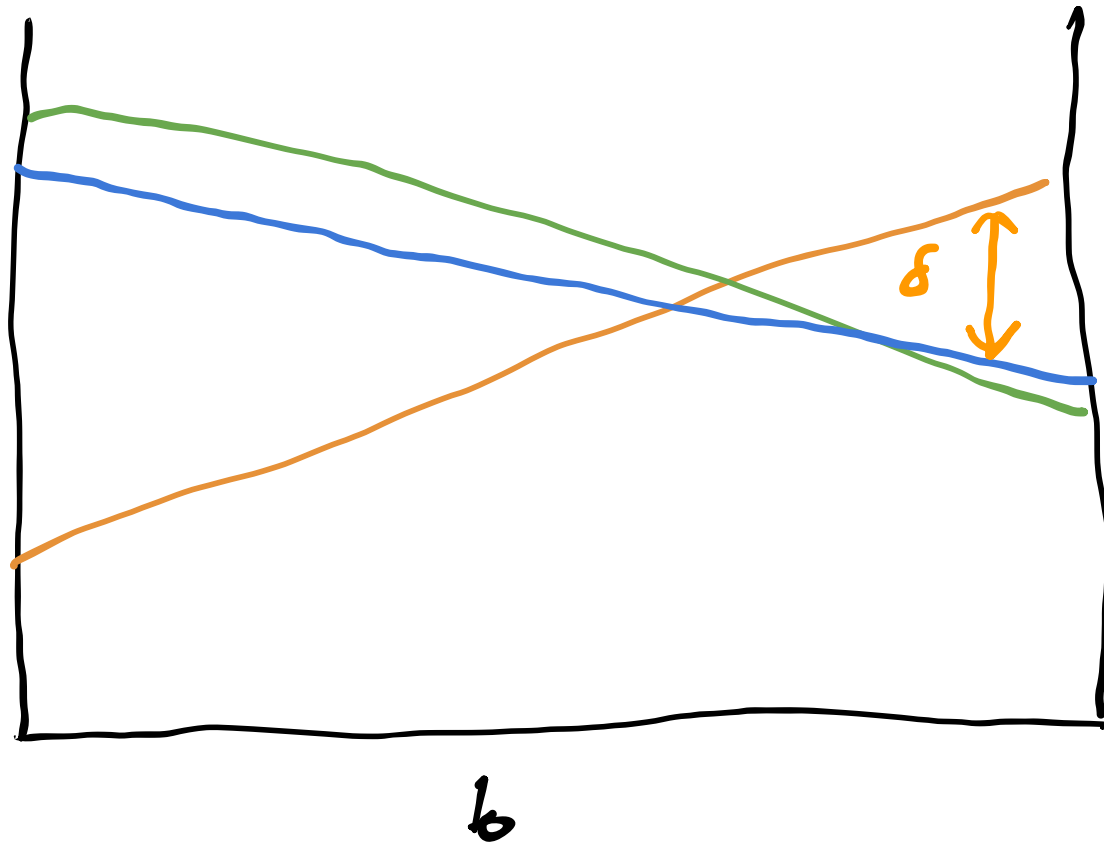
maximize δ
subject to $b \geq 0$

α -Vector Pruning



$$\begin{aligned} & \underset{\delta, b}{\text{maximize}} && \delta \\ & \text{subject to} && b \geq 0 \\ & && \mathbf{1}^\top b = 1 \end{aligned}$$

α -Vector Pruning

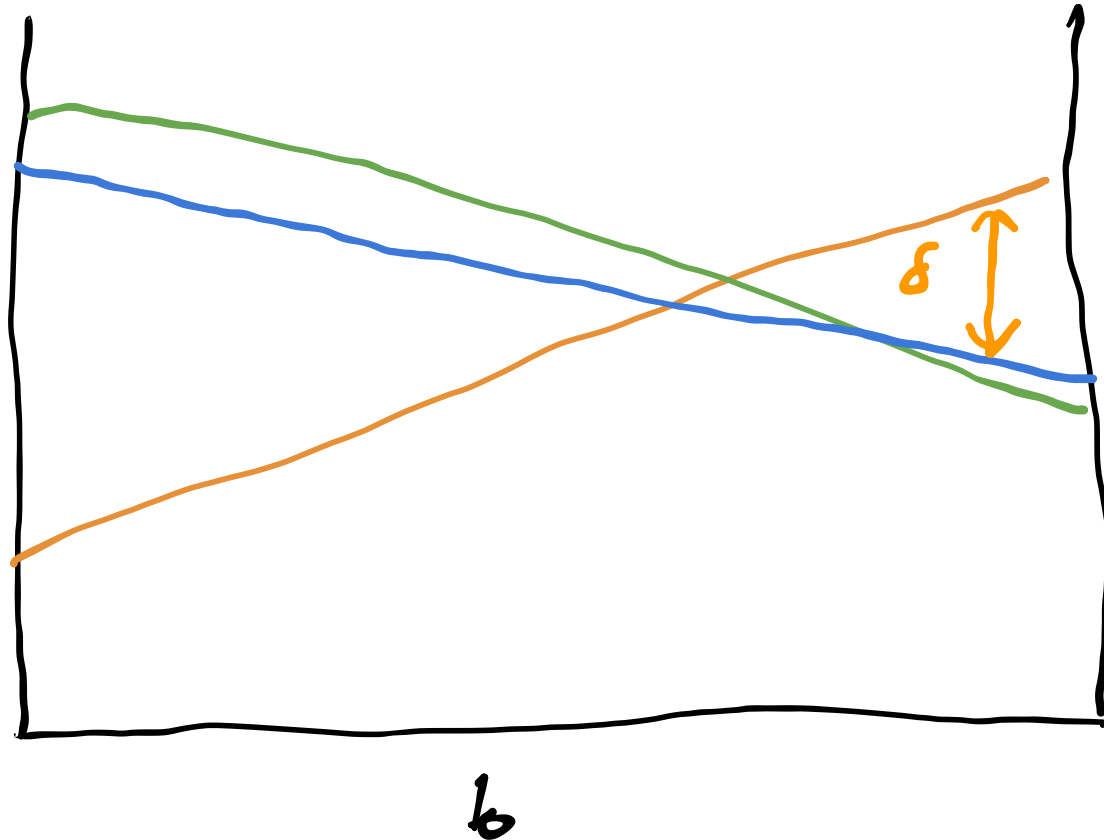


$$\begin{aligned} &\underset{\delta, b}{\text{maximize}} && \delta \\ &\text{subject to} && b \geq 0 \end{aligned}$$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \underline{\alpha'}^\top b + \delta \quad \forall \alpha' \in \underline{\Gamma}$$

α -Vector Pruning



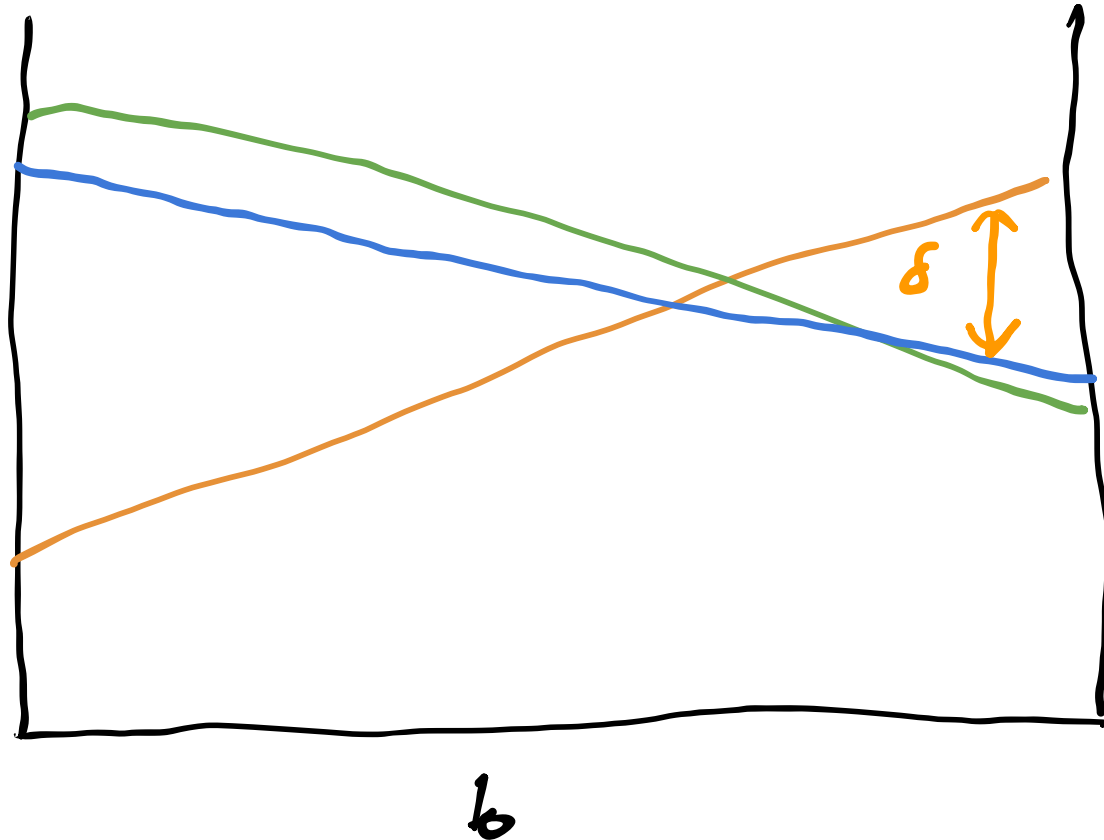
maximize δ
subject to $b \geq 0$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma$$

"Linear Program": easy
optimization problem

α -Vector Pruning



$$\underset{\delta, b}{\text{maximize}} \quad \delta$$

$$\text{subject to} \quad b \geq 0$$

$$\mathbf{1}^\top b = 1$$

$$\alpha^\top b \geq \alpha'^\top b + \delta \quad \forall \alpha' \in \Gamma$$

"Linear Program": easy optimization problem

If $\delta^* > 0$, α is not dominated; b solution sometimes called "witness".

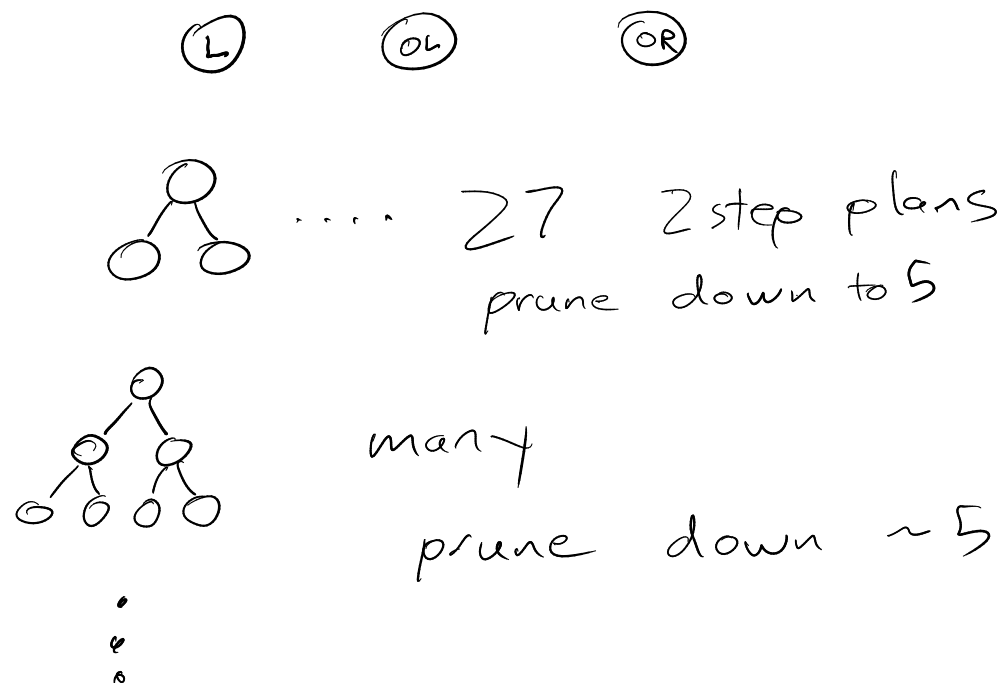
POMDP Value Iteration (horizon d)

$\Gamma^0 \leftarrow \emptyset$

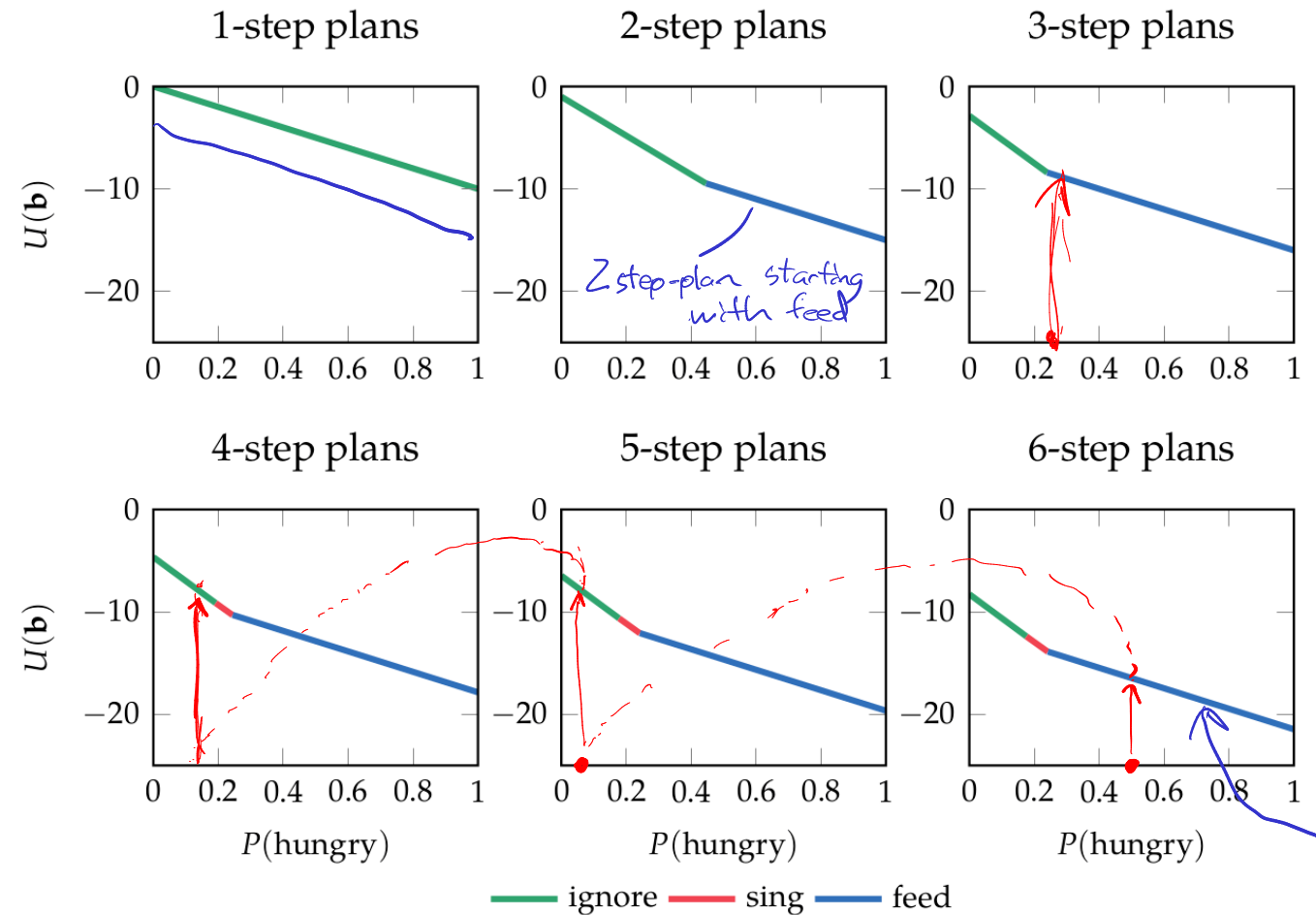
for $n \in 1 \dots d$

Construct Γ^n by expanding with Γ^{n-1}

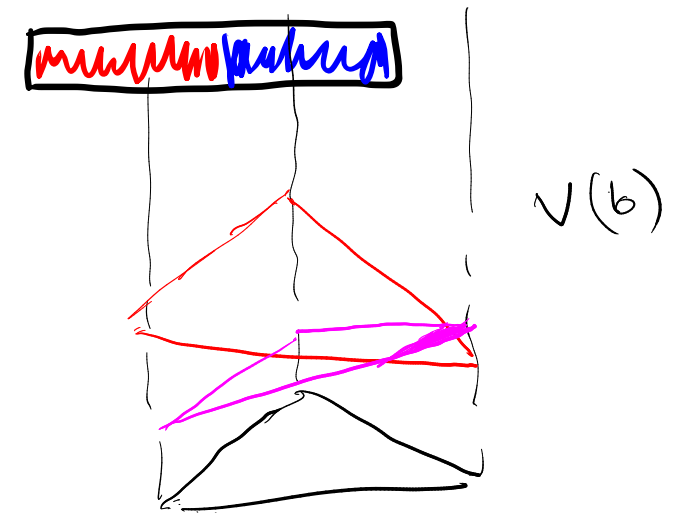
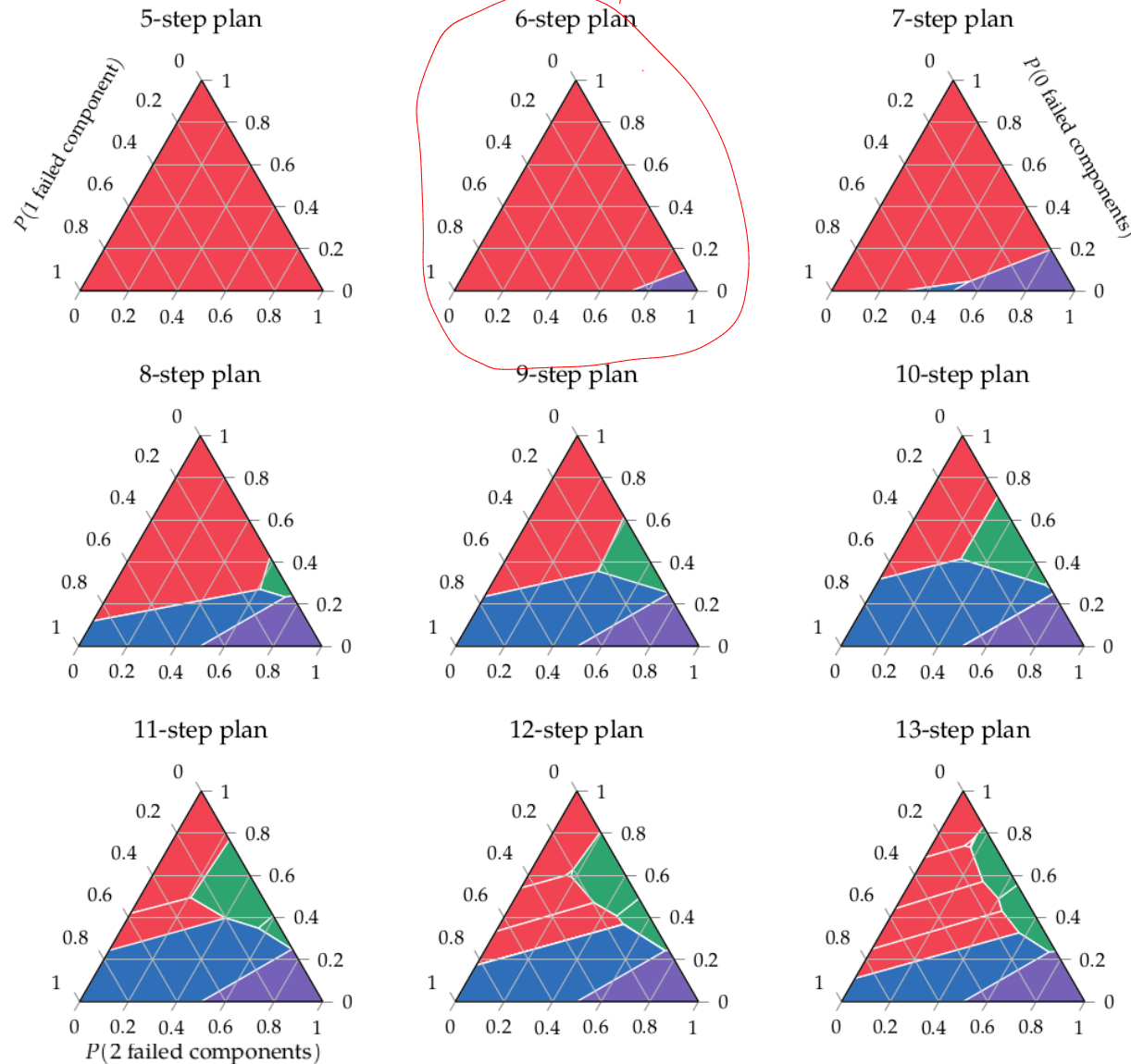
Prune Γ^n



Finite Horizon POMDP Value Iteration



Finite Horizon POMDP Value Iteration



Recap

Recap

- A POMDP is an MDP on the _____

Recap

- A POMDP is an MDP on the belief space

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a _____

Recap

- A POMDP is an MDP on the belief space
- The value function of a discrete POMDP can be represented by a set of α -vectors
- Each α vector corresponds to a conditional plan